

# Faiyke RMM System

Technical Guide

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Architecture, data models, API reference, and extension guide

Version 1.6 · September 2026

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Prepared for

**Faiyke-AI Agency**

# RMM System — Technical Guide

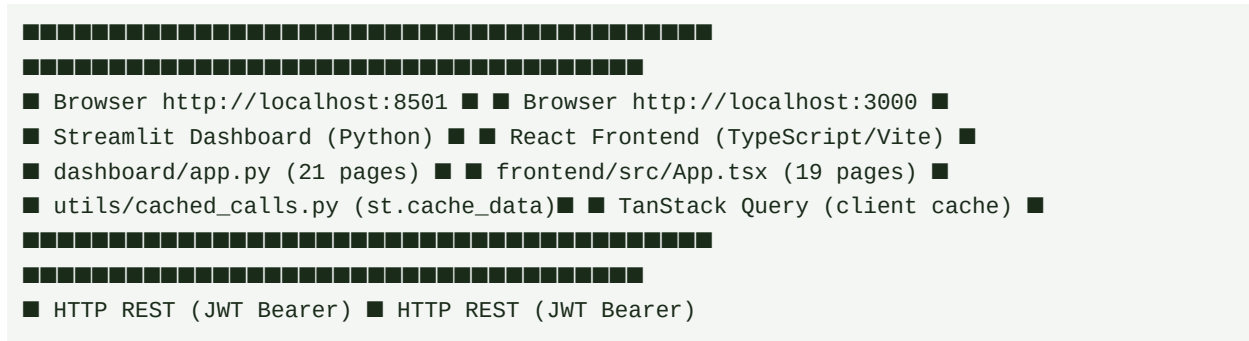
**Audience:** Developers, system architects, and advanced administrators **Stack:** Flask 3 · SQLAlchemy 2 · Celery 5 · Streamlit 1.58 · React 19/Vite 8/TypeScript · PostgreSQL 15 · Redis/Memurai **Version:** 1.5 (React frontend, cross-platform agent, screenshot pipeline, IoT/MQTT, performance hardening, Android MDM, AI Assistant services refactor, API/token usage monitoring)

## Table of Contents

- 1. System Architecture
- 2. Data Models
- 3. API Endpoints
- 4. Authentication & Authorization
- 5. Celery Tasks & Schedules
- 6. Agent Internals
- 7. Built-in Scripts Reference
- 8. Dashboard API Client
- 9. Performance Optimizations
- 10. Extension Guide
- 11. Security Model
- 12. Environment Variables Reference
- 13. Docker Deployment
- 14. AI Assistant
- 15. React Frontend
- 16. Screenshot Capture Pipeline
- 17. IoT / MQTT / SNMP
- 18. Test Suite
- 19. Mobile Device Management
- 20. API & Token Usage Monitoring
- 21. Consistent List Pagination
- 22. Security Audit Remediation & Dependency Hygiene

## 1. System Architecture

### Component Diagram



■ 3-attempt retry + 401 auto-refresh ■ axios + React Query

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[illegible]

■ Flask API <http://localhost:5000> ■

```
■ api/app.py (Sentry init, JWT decode cache, cache pre-warm) ■
```

■ api/routes/\*.py (21 blueprints incl. sensors, terminal) ■

```
■ api/models/*.py (SQLAlchemy models incl. SensorReading) ■
```

- `api/utils/cache.py` (Redis JSON cache + raw Response cache) ■

```
■ api/utils/jwt_cache.py (TTLCache 512 tokens, 60s – skips HMAC) ■
```

```
■ api/utils/oui.py (MAC vendor lookup, 500+ entries) ■
```

- flask-compress (gzip all JSON responses >500B) ■

```
■ waitress (16 threads) ■
```

12345678910111213141516171819202122232425262728293031323334353637383940414243444546474849505152535455565758596061626364656667686970717273747576777879808182838485868788899091929394959697989910010110210310410510610710810911011111211311411511611711811912012112212312412512612712812913013113213313413513613713813914014114214314414514614714814915015115215315415515615715815916016116216316416516616716816917017117217317417517617717817918018118218318418518618718818919019119219319419519619719819920020120220320420520620720820921021121221321421521621721821922022122222322422522622722822923023123223323423523623723823924024124224324424524624724824925025125225325425525625725825926026126226326426526626726826927027127227327427527627727827928028128228328428528628728828929029129229329429529629729829930030130230330430530630730830931031131231331431531631731831932032132232332432532632732832933033133233333433533633733833934034134234334434534634734834935035135235335435535635735835936036136236336436536636736836937037137237337437537637737837938038138238338438538638738838939039139239339439539639739839940040140240340440540640740840941041141241341441541641741841942042142242342442542642742842943043143243343443543643743843944044144244344444544644744844945045145245345445545645745845946046146246346446546646746846947047147247347447547647747847948048148248348448548648748848949049149249349449549649749849950050150250350450550650750850951051151251351451551651751851952052152

■ ■ ■

PostgreSQL Redis (broker) Agent API endpoints

```
:5432 :6379 / PgBouncer /api/agents/*
```

```
db: rmmdb :6432 (optional) X-Agent-Token auth
```

■ ■

[illegible]

■ ■ Celery Worker (--pool=solo on Windows) ■

```
■ ■ tasks/alert_tasks.py – beat 60s/180s ■
```

■ ■ tasks/patch\_tasks.py – beat 1800s ■

■ ■ tasks/automation\_tasks.py – on demand ■

■ ■ tasks/report\_tasks.py – on demand ■

■ ■ tasks/network\_tasks.py – beat 300s ■

```
■ ■ tasks/backup_tasks.py – beat 86400s ■
```

■ ■ tasks/billing\_tasks.py – beat 86400s ■

■ ■ tasks/mqtt\_tasks.py – MQTT subscription ■

■ ■ tasks/snmp\_tasks.py – SNMP polling ■

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## ■ Python Agent (Windows / macOS / Linux) ■

```
■ agent/rmm_agent.py – main loop (60s heartbeat, screenshot) ■
```

■ agent/collector.py – cross-platform metrics/software/patches ■

```
■ agent/heartbeat.py – APIClient (incl. send_screenshot) ■
```

■ agent/executor.py – task execution engine ■

```
■ agent/script_runner.py – ps1/bat/py/sh execution ■
```

■ agent/screenshot.py – PIL/scrot screen capture (300s interval)■

■ agent/terminal\_worker.py – WebSocket terminal bridge ■

■ agent/iot\_agent.py – IoT sensor readings (hwmon/DHT/BME680) ■

■ agent/build.py – PyInstaller single-file binary builder ■

12345678910111213141516171819202122232425262728293031323334353637383940414243444546474849505152535455565758596061626364656667686970717273747576777879808182838485868788899091929394959697989910010110210310410510610710810911011111211311411511611711811912012112212312412512612712812913013113213313413513613713813914014114214314414514614714814915015115215315415515615715815916016116216316416516616716816917017117217317417517617717817918018118218318418518618718818919019119219319419519619719819920020120220320420520620720820921021121221321421521621721821922022122222322422522622722822923023123223323423523623723823924024124224324424524624724824925025125225325425525625725825926026126226326426526626726826927027127227327427527627727827928028128228328428528628728828929029129229329429529629729829930030130230330430530630730830931031131231331431531631731831932032132232332432532632732832933033133233333433533633733833934034134234334434534634734834935035135235335435535635735835936036136236336436536636736836937037137237337437537637737837938038138238338438538638738838939039139239339439539639739839940040140240340440540640740840941041141241341441541641741841942042142242342442542642742842943043143243343443543643743843944044144244344444544644744844945045145245345445545645745845946046146246346446546646746846947047147247347447547647747847948048148248348448548648748848949049149249349449549649749849950050150250350450550650750850951051151251351451551651751851952052152

## Request Flow: Dashboard → API → Agent

```
User clicks "Reboot" in Maintenance page
■
▼
dashboard/pages/15_Maintenance.py
■■ client.reboot_device(device_id)
■■ POST /api/devices/<id>/reboot
■■ routes/devices.py: reboot_device()
■■ _queue_built_in_task(device_id, "reboot")
■■ ScriptRun(script_id=<builtin_reboot_id>, device_id=...)
■■ db.session.add + commit
■
(60s later)
■
▼
agent polls GET /api/agents/<id>/tasks
■■ returns [{type: "run_script", script_content: "..."}]
■■ executor.run(task)
■■ script_runner.run_ps1("Restart-Computer -Force")
```

## 2. Data Models

### Model Relationships

```
Customer ■■< Device ■■< DeviceMetrics
■ ■■■< InstalledSoftware
■ ■■■< AlertRule.customer_id (optional)
■ ■■■< PatchRecord
■ ■■■< ScriptRun
■ ■■■< ScheduledTaskRun
■
■■■< Ticket ■■< TicketComment
■ ■■■ assignee_id → User
■
■■■< PatchPolicy
■■■< AutomationProfile ■■< ScheduledTaskRun
■■■< Invoice
■■■< Report
■■■< NetworkScan

AlertRule ■■< Alert
Script ■■< ScriptRun
User ■■< AuditLog
Device ■■< AgentToken
```

## Model Reference

### User (users)

```
id Integer PK
email String(255) UNIQUE NOT NULL
password_hash String(255) # bcrypt rounds=12
full_name String(255)
role String(50) # superadmin | admin | technician | viewer
# superadmin: highest privilege, bypasses all _require_role() checks,
# cannot be deleted or modified via API
mfa_secret String(32)
mfa_enabled Boolean default False
is_active Boolean default True
must_change_password Boolean default False server_default="false"
# True → intercept login with force-change screen
created_at DateTime server_default=now()
```

### Customer (customers)

```
id Integer PK
name String(255) NOT NULL
slug String(100) UNIQUE
email String(255)
phone String(50)
tier String(50) default 'standard' # standard | premium | enterprise
notes Text
is_active Boolean default True
created_at DateTime
```

#### Device (devices)

```
id String/UUID PK
customer_id FK → customers
group_id FK → device_groups (nullable)
hostname String(255)
platform String(50) # windows | linux | mac
os_name String(255)
os_version String(100)
cpu_brand String(255)
cpu_cores Integer
ram_gb Float
ip_address String(50) # local IP
serial_number String(255)
hardware_id String(64) UNIQUE # SHA256(hostname+MAC)
status String(50) default 'active'
is_online Boolean default False
last_seen DateTime
agent_version String(20)
display_name String(255)
created_at DateTime
is_agentless Boolean default False NOT NULL # True = network-discovered, no agent
device_type String(50) default 'laptop' NOT NULL # laptop | desktop | mobile | server | unknown
vendor String(255) nullable # OUI lookup result, e.g. "Apple, Inc."
# NOTE: customer_id is nullable=True – agentless devices start unassigned
```

#### DeviceMetrics (device\_metrics)

```
id Integer PK
device_id FK → devices
cpu_pct Float
ram_pct Float
disk_pct Float # primary disk
battery_pct Float (nullable)
uptime_seconds BigInteger
top_processes JSON # [{pid, name, cpu}]
disks JSON # [{mountpoint, device, total_gb, used_gb, free_gb, percent}]
collected_at DateTime # NOTE: field is collected_at not recorded_at
```

#### AlertRule (alert\_rules)

```
id Integer PK
name String(255) NOT NULL
metric String(100) # cpu_pct | ram_pct | disk_pct | offline
operator String(10) # > | < | ==
threshold Float
severity String(20) # info | warning | critical
cooldown_minutes Integer default 15
is_active Boolean default True
customer_id FK → customers (nullable = global rule)
notification_channels JSON # {"email": ["addr@domain.com"]}
```

## Alert (alerts)

```
id Integer PK
rule_id FK → alert_rules
device_id FK → devices
severity String(20)
status String(20) default 'open' # open | acknowledged | resolved
message Text
acknowledged_by FK → users (nullable)
created_at DateTime
resolved_at DateTime (nullable)
```

## Ticket + TicketComment

```
# Ticket
id String/UUID PK
title String(500) NOT NULL
description Text
customer_id FK → customers
device_id FK → devices (nullable)
assignee_id FK → users (nullable) # Phase 8 addition
priority String(20) default 'medium' # low | medium | high | critical
status String(20) default 'open' # open | in_progress | resolved | closed
source String(50) # manual | alert | agent
created_at DateTime
updated_at DateTime onupdate=now()

# TicketComment
id Integer PK
ticket_id FK → tickets
author_id FK → users
body Text NOT NULL
is_internal Boolean default False
created_at DateTime
```

#### Script + ScriptRun

```
# Script
id String/UUID PK
name String(255) NOT NULL
description Text # used as __tag__ for built-in scripts
file_type String(20) # ps1 | bat | py | sh
content Text NOT NULL
os_target String(20) default 'windows'
is_builtin Boolean default False
created_by FK → users (nullable)
created_at DateTime

# ScriptRun
id String/UUID PK
script_id FK → scripts NOT NULL
device_id FK → devices
triggered_by FK → users (nullable)
exit_code Integer
stdout Text
stderr Text
status String(20) default 'pending' # pending | running | completed | failed | timeout
timeout_seconds Integer default 300
started_at DateTime (nullable)
completed_at DateTime (nullable)
created_at DateTime
```

#### PatchPolicy + PatchRecord

```
# PatchPolicy
id Integer PK
name String(255) NOT NULL
customer_id FK → customers (nullable)
auto_approve_critical Boolean default True
auto_approve_important Boolean default False
auto_approve_security Boolean default False
reboot_behavior String(50) default 'prompt' # prompt | auto | never
maintenance_window_start String(10) # HH:MM
maintenance_window_end String(10)
excluded_software JSON # list of software name patterns to skip
is_active Boolean default True

# PatchRecord
id Integer PK
device_id FK → devices
policy_id FK → patch_policies (nullable)
patch_name String(500)
kb_id String(50)
patch_type String(50) # critical | security | definition | update | feature | driver
status String(50) default 'pending' # pending | approved | deployed | failed
deployed_at DateTime (nullable)
created_at DateTime
```

#### AutomationProfile + ScheduledTaskRun

```
# AutomationProfile
id Integer PK
name String(255) NOT NULL
customer_id FK → customers (nullable)
schedule_type String(50) # daily | weekly | monthly | once
schedule_day String(20) # monday..sunday or day-of-month
schedule_time String(10) # HH:MM
os_patch_config JSON # {install_all, critical, security, definition, ...}
software_patch_config JSON # {update_all, excluded: [...]}
disk_config JSON # {defrag: bool, checkdisk: bool}
maintenance_config JSON # {delete_temp, restore_point, clear_history, reboot, shutdown}
notification_emails String(500)
run_on_new_agents Boolean default False
is_active Boolean default True
last_run_at DateTime (nullable)
created_at DateTime

# ScheduledTaskRun
id Integer PK
profile_id FK → automation_profiles
device_id FK → devices
status String(20) default 'pending'
result_summary JSON
started_at DateTime (nullable)
finished_at DateTime (nullable)
created_at DateTime
```

#### Report

```
id Integer PK
name String(255)
template_type String(100) # device_health | patch_compliance | alert_summary |
software_inventory | ticket_summary
customer_id FK → customers (nullable)
format String(10) default 'csv'
file_path String(500) # absolute path to generated CSV in api/reports/
generated_by FK → users (nullable)
created_at DateTime

# IMPORTANT: to_dict() must include "file_path" for dashboard download to work
```

#### Invoice (invoices)

```
id String/UUID PK
customer_id FK → customers
invoice_number String(20) UNIQUE nullable index # INV-YYYY-NNNN, auto-generated on create
period_start DateTime(timezone=True) NOT NULL
period_end DateTime(timezone=True) NOT NULL
due_date DateTime(timezone=True) nullable # defaults to period_end + 30 days
device_count Integer
per_device_rate Numeric(10,2) NOT NULL
tax_rate Numeric(5,4) default 0.0 # stored as decimal: 0.20 = 20%
notes Text nullable # per-invoice custom notes
subtotal Numeric(10,2) # device_count × per_device_rate
tax_amount Numeric(10,2) # subtotal × tax_rate
total Numeric(10,2) # subtotal + tax_amount
status String(20) default 'draft' # draft | sent | paid | overdue
sent_at DateTime nullable
paid_at DateTime nullable
created_at DateTime
```

#### Sequential invoice number generation (api/routes/billing.py:\_next\_invoice\_number()):

```
year = datetime.now(timezone.utc).year
count = Invoice.query.filter(Invoice.invoice_number.like(f"INV-{year}-%")).count()
return f"INV-{year}-{count + 1:04d}"
```

#### OrgSettings (org\_settings)

Singleton model — always id=1. Seeded at API startup by ensure\_org\_settings().

```
id Integer PK default 1 company_name String(255) default "" company_address Text default ""
company_email String(255) default "" company_phone String(50) default "" logo_data Text nullable #
base64 data URI (same pattern as user avatar) payment_terms String(100) default "Net 30" bank_details
Text default "" footer_notes Text default "Thank you for your business!"
```

## White-label branding

```
app_name String(100) default "RMM System" tagline String(200) default "Remote Monitoring &
Management" primary_color String(7) default "#407E3C"
```

## Regional settings (added migration f1a2b3c4d5e6, down\_revision e6f7a8b9c0d1)

```
currency String(3) nullable server_default='USD' # ISO 4217 code timezone String(50) nullable
server_default='UTC' # IANA tz name
```

``logo_bytes()`` method decodes the base64 data URI and returns raw bytes for use in PDF generation.

``ensure_org_settings()`` is called from ``create_app()`` — safe to call multiple times (no-op if row exists).

``to_dict()`` returns ``currency`` (default ``"USD"``) and ``timezone`` (default ``"UTC"``)

alongside all other fields. Dashboard reads these from  
`st.session\_state["\_org\_settings"]` (loaded once per session in `require\_auth()`).

---

### ## 3. API Endpoints

All endpoints prefixed with `/api/`. Protected by `@jwt\_required()` unless noted.

#### ### Auth (`/api/auth/`)

| Method | Path                      | Auth          | Description                                                                                                                           |
|--------|---------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| POST   | /login                    | None          | Email+password → access+refresh tokens, OR `{"status": "mfa_required", "mfa_token": "...}"` (HTTP 200) if user has `mfa_enabled=True` |
| POST   | /refresh                  | Refresh token | Rotate access token                                                                                                                   |
| POST   | /logout                   | JWT           | Revoke token                                                                                                                          |
| POST   | /me/force-change-password | JWT           | Set new password + clear `must_change_password` flag                                                                                  |
| PUT    | /me/password              | JWT           | Change password (current + new). Used by Profile page.                                                                                |
| POST   | /mfa/setup                | JWT           | Generate provisional TOTP secret. Returns `{secret, provisioning_uri}`. Not active until /mfa/enable called.                          |
| POST   | /mfa/enable               | JWT           | Activate MFA. Body: `{code}`. Requires valid TOTP code.                                                                               |
| POST   | /mfa/login                | None          | Second login step. Body: `{mfa_token, code}`. Returns full JWT. Rate-limited 10/min.                                                  |
| POST   | /mfa/disable              | JWT           | Disable MFA. Body: `{password}`. Requires current password.                                                                           |

#### ### Agents (`/api/agents/`)

| Method | Path                     | Auth             | Description                                                            |
|--------|--------------------------|------------------|------------------------------------------------------------------------|
| POST   | /register                | None (org_token) | Register new agent, get device_id + agent_token. Rate-limited: 10/min. |
| POST   | /<device_id>/heartbeat   | X-Agent-Token    | Submit metrics, get queued tasks back                                  |
| GET    | /<device_id>/tasks       | X-Agent-Token    | Poll pending ScriptRun records                                         |
| POST   | /<device_id>/task_result | X-Agent-Token    | Submit ScriptRun result                                                |
| PUT    | /<device_id>/patches     | X-Agent-Token    | Submit pending Windows patches list                                    |
| PUT    | /<device_id>/software    | X-Agent-Token    | Submit installed software list                                         |

#### ### Devices (`/api/devices/`)

| Method | Path                 | Auth           | Description                                                                           |
|--------|----------------------|----------------|---------------------------------------------------------------------------------------|
| GET    | /                    | JWT            | List devices (paginated, filterable)                                                  |
| GET    | /<id>                | JWT            | Get device + latest metrics                                                           |
| PUT    | /<id>                | JWT admin/tech | Update display_name, group_id                                                         |
| DELETE | /<id>                | JWT admin      | Remove device                                                                         |
| GET    | /<id>/metrics        | JWT            | Historical metrics (up to 168h). Row cap: `.limit(5000)` prevents unbounded response. |
| GET    | /<id>/software       | JWT            | Installed software list                                                               |
| POST   | /<id>/reboot         | JWT admin/tech | Queue reboot task                                                                     |
| POST   | /<id>/shutdown       | JWT admin/tech | Queue shutdown task                                                                   |
| POST   | /<id>/queue_task     | JWT admin/tech | Queue any built-in maintenance task                                                   |
| POST   | /<id>/deploy_patches | JWT admin/tech | Trigger patch deployment via Celery                                                   |
| GET    | /platform_counts     | JWT            | Returns `{"by_platform": {windows: N, ...},`                                          |

```

"agentless": N} ` |
| POST | `<id>/ping_check` | JWT admin/tech | Ping device IP, update
is_online/last_seen. Only for is_agentless=True devices. |

**`queue_task` valid `task_type` values:**
`clean_temp`, `defrag`, `check_disk`, `restore_point`, `clear_browser`, `reboot`,
`shutdown`

### Alerts (`/api/alerts/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List alerts (filter by status, severity) |
| GET | `/rules` | JWT | List alert rules |
| POST | `/rules` | JWT admin/tech | Create alert rule |
| PUT | `/rules/<id>` | JWT admin/tech | Update alert rule |
| DELETE | `/rules/<id>` | JWT admin | Delete alert rule |
| POST | `<id>/acknowledge` | JWT | Acknowledge alert |
| POST | `<id>/resolve` | JWT | Resolve alert |

### Tickets (`/api/tickets/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List tickets (filter by status, priority) |
| POST | `/` | JWT tech/admin | Create ticket (was unguarded – any JWT; now requires
`technician` or `admin` role) |
| GET | `<id>` | JWT | Get ticket + comments |
| PUT | `<id>` | JWT tech/admin | Update status, priority, assignee_id |
| DELETE | `<id>` | JWT admin | Delete ticket |
| POST | `<id>/comments` | JWT | Add comment |

### Scripts (`/api/scripts/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List scripts (filter built-in/custom) |
| POST | `/` | JWT tech/admin | Create script |
| PUT | `<id>` | JWT tech/admin | Update script |
| DELETE | `<id>` | JWT admin | Delete script |
| POST | `<id>/run` | JWT tech/admin | Create ScriptRun records for target devices |
| GET | `/runs` | JWT | List recent ScriptRun records |
| GET | `/runs/<id>` | JWT | Get ScriptRun detail (stdout, stderr) |

### Patches (`/api/patches/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List patch records (filter by status, device) |
| PUT | `<id>/approve` | JWT tech/admin | Approve a patch record |
| POST | `/approve_bulk` | JWT tech/admin | Approve multiple patch records |
| GET | `/policies` | JWT | List patch policies |
| POST | `/policies` | JWT admin | Create patch policy |
| PUT | `/policies/<id>` | JWT admin | Update patch policy |

### Automation (`/api/automation/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|

```

```
| GET | `/profiles` | JWT | List automation profiles |
| POST | `/profiles` | JWT admin/tech | Create profile |
| PUT | `/profiles/<id>` | JWT admin/tech | Update profile |
| DELETE | `/profiles/<id>` | JWT admin | Delete profile |
| POST | `/profiles/<id>/run` | JWT admin/tech | Trigger immediate profile run |
| GET | `/runs` | JWT | List ScheduledTaskRun records |
```

#### ### Reports (`/api/reports/`)

```
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List reports |
| POST | `/` | JWT admin/tech | Create report + trigger generation |
| GET | `/<id>` | JWT | Get report metadata (includes file_path) |
```

#### ### Billing (`/api/billing/`)

```
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/invoices` | JWT | List invoices (optional `?customer_id=`) |
| POST | `/invoices` | JWT admin | Generate invoice – auto-assigns `invoice_number` (INV-YYYY-NNNN), accepts `tax_rate`, `due_date`, `notes` |
| GET | `/invoices/<id>` | JWT | Get single invoice |
| PUT | `/invoices/<id>/status` | JWT admin | Update invoice status (draft→sent→paid/overdue) |
| DELETE | `/invoices/<id>` | JWT admin | Delete invoice |
| GET | `/invoices/<id>/pdf` | JWT | Generate and stream A4 PDF (Content-Disposition: attachment) |
| POST | `/invoices/<id>/send-email` | JWT admin | Email PDF to customer via SMTP. Returns 503 if `SMTP_HOST` not configured. |
```

#### ### Org Settings (`/api/admin/`)

```
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/org-settings` | JWT admin | Return OrgSettings singleton (includes `logo_data` base64 URI, `currency`, `timezone`) |
| PUT | `/org-settings` | JWT admin | Update text fields: company name, address, email, phone, payment terms, bank details, footer notes, `app_name`, `tagline`, `primary_color`, **`currency`** (ISO 3-letter), **`timezone`** (IANA name) |
| PUT | `/org-settings/logo` | JWT admin | Upload logo – multipart file upload, Pillow-resized to max 400px wide, stored as base64 |
| DELETE | `/org-settings/logo` | JWT admin | Remove logo (sets `logo_data` = null) |
```

#### ### Customers (`/api/customers/`)

```
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/` | JWT | List customers |
| POST | `/` | JWT admin/tech | Create customer |
| GET | `/<id>` | JWT | Get customer detail |
| PUT | `/<id>` | JWT admin/tech | Update customer |
| DELETE | `/<id>` | JWT admin | Deactivate customer |
```

#### ### Users (`/api/users/`)

```
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
```

```

| GET | `/` | JWT admin | List users |
| POST | `/` | JWT admin | Create user |
| PUT | `/<id>` | JWT admin | Update user (role, active, password) |

### Admin (`/api/admin/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/users` | JWT admin | List all users |
| POST | `/users` | JWT admin | Create user – accepts `must_change_password` bool |
| PUT | `/users/<id>` | JWT admin | Update user – accepts `must_change_password` bool.
Returns 403 if target is superadmin. |
| DELETE | `/users/<id>` | JWT admin | Delete user. Returns 403 if target is superadmin.
|
| GET | `/org-token` | JWT admin | Return `ORG_REGISTRATION_TOKEN` for display in Admin
panel |
| GET | `/server_ips` | None | Returns `{"hostname": "...", "lan_ips": ["192.168.x.x"]}`
via socket.getaddrinfo |

### Network (`/api/network/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| POST | `/scan` | JWT admin/tech | Trigger network discovery scan (dispatches
`run_network_scan.delay`) |
| GET | `/scans` | JWT | List scan results |
| POST | `/agentless_devices` | JWT admin/tech | Batch upsert agentless devices. Body:
`{"hosts": [...], "customer_id": optional}`. Returns `{"created": N, "updated": M,
"skipped": K}`. |

### Dashboard Summary (`/api/dashboard/`)
| Method | Path | Auth | Description |
|-----|-----|-----|-----|
| GET | `/summary` | JWT | Aggregated counts (devices, alerts, tickets, online) |
| GET | `/health_map` | JWT | Device health list (limit 500) – returned objects include
`id`, `hostname`, `status`, `is_online`; used by Dashboard Overview to render clickable
cards |
| GET | `/activity` | JWT | Recent audit log entries |

---

## 4. Authentication & Authorization

### JWT Flow

```

Login (no MFA): POST /api/auth/login {email, password} → {access\_token, refresh\_token, user: {id, email, role, full\_name, must\_change\_password}}

Login (MFA enabled): POST /api/auth/login {email, password} → HTTP 200 {status: "mfa\_required", mfa\_token: "<short-lived JWT>"} mfa\_token is a 5-min TTL JWT with claim {purpose: "mfa\_pending"}. Cannot be used for any other API call. Client must immediately call: POST /api/auth/mfa/login {mfa\_token, code} → {access\_token, refresh\_token, user: {...}}

Force password change (if `must_change_password=True`): Dashboard intercepts in `app.py` before showing any page. User fills new password form → `POST /api/auth/me/force-change-password {new_password}` → clears `must_change_password` flag → `st.rerun()` → normal dashboard

Dashboard stores: `st.session_state["access_token"] = access_token` `st.session_state["refresh_token"] = refresh_token` `st.session_state["user"] = user dict` `st.query_params["tok"] = access_token` ← written once at login `st.query_params["rtok"] = refresh_token` ← written once at login (Tokens are stripped from URL by `require_auth()` after first restore. Session state carries them from that point. Mitigates browser history exposure.)

API call: Authorization: Bearer <access\_token>

On 401: `POST /api/auth/refresh` (Authorization: Bearer <refresh\_token>) → {access\_token} Update `session_state`, retry original request once

On page reload (F5) or navigation: Streamlit session state is wiped on browser reload. `app.py` reads `st.query_params["tok"]` and `["rtok"]` before session state check. `require_auth()` in every page also re-reads and re-stamps both params. `require_auth()` re-stamps `?tok=` and `?rtok=` on every authenticated page load — ensures F5 on any page restores both access and refresh tokens, allowing silent renewal even after the 900s access token TTL expires.

```
> **NOTE:** Both access (`?tok=`) and refresh (`?rtok=`) tokens are stored in URL params.
> This is a pragmatic trade-off for Streamlit's stateless page reload model – without it,
> any F5 reload logs the user out. Mitigated by short access token TTL (900s default).
> In production with HTTPS, tokens in URL are not visible in transit but appear in server
> access logs – rotate `JWT_ACCESS_TOKEN_EXPIRES` to 300s for higher-security
environments.

### MFA Flow
```

Setup: `POST /api/auth/mfa/setup` → {secret, provisioning\_uri} (provisional -- not active yet)

Enable (after QR scan): `POST /api/auth/mfa/enable {code: "123456"}` → {message: "MFA enabled"} -- sets `mfa_enabled=True` on User

Login with MFA: `POST /api/auth/login {email, password}` → {status: "mfa\_required", mfa\_token: "<5-min JWT>"} (HTTP 200) `mfa_token` JWT claims: {purpose: "mfa\_pending", sub: user\_id} Cannot be used for any endpoint except `/mfa/login`.

`POST /api/auth/mfa/login {mfa_token, code}` Rate-limited: 10/min → {access\_token, refresh\_token, user: {...}} (full session)

Disable: `POST /api/auth/mfa/disable {password}` → {message: "MFA disabled"} -- requires current password for confirmation

```
**Dashboard flow:**
- `login()` in `dashboard/utils/auth.py` returns `"ok"` / `"mfa_required"` / `"error"`
(not bool).
- On `mfa_required`: stores `mfa_token` in `st.session_state["mfa_pending_token"]`.
- `app.py` route block checks `mfa_pending_token` before `access_token` – shows
```

```
`show_mfa_step()` full-screen form.
- Back button clears `mfa_pending_token` and returns to login.

**QR code generation:** `qrcode[pil]==8.0` in `dashboard/requirements.txt`.

---

### Agent Token Flow
```

Agent registers: POST /api/agents/register Body: {hardware\_id, hostname, platform, os\_name, cpu\_brand, cpu\_cores, ram\_gb, ip\_local, mac\_address, org\_token} → {device\_id, agent\_token}

Stored in: agent/agent\_state.json

Subsequent requests: Header: X-Agent-Token: <agent\_token>

Token stored hashed in agent\_tokens table. On 401: agent re-registers (full registration flow).

#### ### Role Permissions Matrix

| Action                           | viewer | technician | admin | superadmin   |
|----------------------------------|--------|------------|-------|--------------|
| View dashboard, devices, alerts  | ✓      | ✓          | ✓     | ✓            |
| Create/update tickets            | –      | ✓          | ✓     | ✓            |
| Acknowledge/resolve alerts       | –      | ✓          | ✓     | ✓            |
| Run scripts                      | –      | ✓          | ✓     | ✓            |
| Queue maintenance tasks          | –      | ✓          | ✓     | ✓            |
| Approve/deploy patches           | –      | ✓          | ✓     | ✓            |
| Manage automation profiles       | –      | ✓          | ✓     | ✓            |
| Create/update customers          | –      | ✓          | ✓     | ✓            |
| Generate reports                 | –      | ✓          | ✓     | ✓            |
| Create/manage users              | –      | –          | ✓     | ✓            |
| Delete any resource              | –      | –          | ✓     | ✓            |
| View Admin page                  | –      | –          | ✓     | ✓            |
| Manage billing/invoices          | –      | –          | ✓     | ✓            |
| Modify/delete superadmin account | –      | –          | –     | – (CLI only) |

> **\*\*NOTE:\*\*** All `\_require\_role()` guards in all 9 route files include a superadmin bypass: `if claims.get("role") == "superadmin": return None`. The superadmin account itself cannot be modified or deleted via any API endpoint – use `api/reset\_superadmin.py` for password changes.

---

#### ## 5. Celery Tasks & Schedules

##### ### Beat Schedule

| Task                     | Schedule    | Module                |
|--------------------------|-------------|-----------------------|
| `evaluate_all_rules`     | Every 60s   | `tasks.alert_tasks`   |
| `mark_offline_devices`   | Every 180s  | `tasks.alert_tasks`   |
| `sync_patch_status`      | Every 1800s | `tasks.patch_tasks`   |
| `ping_agentless_devices` | Every 300s  | `tasks.network_tasks` |

```
| `backup_database` | Every 86400s | `tasks.backup_tasks` |  
| `generate_recurring_invoices` | Every 86400s | `tasks.billing_tasks` |
```

### ### Task Signatures

#### #### `evaluate\_all\_rules()`

- Fetches all active alert rules
- **\*\*N+1 fix:\*\*** collects all device IDs per rule batch, then loads all latest metrics in a single subquery:

```
max_id_subq = db.select(func.max(DeviceMetrics.id).label("max_id"))\n.where(...).group_by(DeviceMetrics.device_id).subquery() latest_by_device = {m.device_id: m for m in DeviceMetrics.query.join(max_id_subq, ...).all()}
```

Previously fired one DB query per device per rule (up to 5,000 queries/cycle).

Now: 1 query per rule batch regardless of device count.

- Evaluates `metric operator threshold` (e.g., `cpu\_pct > 90`)
- Respects `cooldown\_minutes` using two separate maps: `open\_alert\_map` (no time filter, prevents re-fire while alert is open) and `resolved\_alert\_map` (recently resolved, for post-recovery cooldown). Fixes storm where alerts older than cooldown\_minutes were evicted and re-fired every 60s indefinitely.
- **\*\*Auto-resolve:\*\*** when metric drops back below threshold, any open alerts for that rule+device are resolved automatically (`status="resolved", `resolved\_at=now()`).
- Creates `Alert` record on breach
- Alert message includes last-seen timestamp for offline alerts:  
`"DESKTOP-ABC has gone offline (last seen: 2024-01-15 14:32 UTC)"`
- Calls `send\_alert\_notification()` if `notification\_channels.email` is set

#### #### `mark\_offline\_devices()`

- Marks devices as `is\_online=False` if `last\_seen` > 5 minutes ago
- Creates offline alerts if a matching rule exists

#### #### `sync\_patch\_status()`

- Iterates all active `PatchPolicy` records
- For each policy's scope (customer or global):
- Auto-approves `PatchRecord` rows where `status='pending'` matching policy flags
- Respects `excluded\_software` name patterns

#### #### `deploy\_patches(device\_id, patch\_ids)`

- Looks up `PatchRecord` rows by ID
- Builds a PowerShell script using PSWindowsUpdate to install those KB IDs
- Creates or updates a `Script` record with tag `\_\_deploy\_patches\_transient\_\_`
- Creates a `ScriptRun` pointing to that script for the device
- Marks `PatchRecord.status = 'deployed'`

#### #### `generate\_report(report\_id)`

- Loads `Report` record by ID
- Dispatches to template-specific data collector:
- `device\_health` – all devices + batch-fetched latest metrics
- `patch\_compliance` – all patch records + device/policy joins
- `alert\_summary` – all alerts + rule names + device hostnames
- `software\_inventory` – all installed software + device hostnames
- `ticket\_summary` – all tickets + customer names
- Writes CSV to `api/reports/<report\_id>\_<name>.csv`

- Updates `report.file\_path` in DB

#### `enqueue\_profile\_run(profile\_id)`

- Loads `AutomationProfile`
- Queries all devices in scope (by `customer\_id` or global)
- Calls `\_dispatch\_profile\_tasks(profile, device\_id, db\_session)` per device
- Updates `profile.last\_run\_at`

#### `\_dispatch\_profile\_tasks(profile, device\_id, db\_session)`

- Reads `disk\_config` and `maintenance\_config` JSON
- For each enabled task type, creates a `ScriptRun` pointing to the corresponding built-in script
- Also creates `ScriptRun` records for any custom scripts attached to the profile

#### `run\_network\_scan(scan\_id)`

- Parses CIDR range from the `NetworkScan` record
- Concurrent ICMP ping sweep via `ThreadPoolExecutor(50)` – all IPs in range pinged in parallel
- For each live IP: ARP table lookup for MAC address, OUI vendor lookup via `\_lookup\_vendor(mac)`
- Reverse DNS via `\_get\_hostname(ip)` runs **before** port probe so hostname is available for all fallbacks
- Detection pipeline (stops at first match):
  1. `\_guess\_platform(vendor)` – OUI vendor string keywords
  2. `\_probe\_platform(ip)` – TCP ports: 62078→iOS, 5555→Android, **2+** of\*\* (445/3389/139)→Windows (2-port threshold prevents routers/NAS with only SMB from being misidentified), 548→macOS, 22→Linux; skipped entirely for router hostnames
  3. `\_guess\_platform\_from\_hostname(rdns)` – 50+ keyword match on rDNS hostname (brand names + Samsung model numbers S10-S24, Note, A12-A73, Fold, Flip, Ultra); catches Android phones with ADB disabled
- Router/gateway hostnames (`fritz.box`, `router`, `gateway`, `modem`, etc.) are included in `discovered\_hosts` for scan display but **never persisted** as agentless device records
- Calls `\_upsert\_agentless\_host(...)` for all non-router live hosts
- Updates `NetworkScan` record with status, host count, and completion timestamp

`\_upsert\_agentless\_host` lookup order (stops at first match):

1. MAC address – primary stable key
  2. IP address across ALL devices (not just agentless) – prevents duplicates for agent-managed PCs
  3. Bare hostname match (strips `.fritz.box`/`.local`, case-insensitive) – catches multi-adapter machines where agent and scan see different MACs (e.g. Ethernet vs WiFi adapter)
- On match with non-agentless device: returns `"skipped"` – never modifies agent-managed records
  - On match with agentless device: upgrades `platform`/`device\_type` if currently "unknown" and new scan detects better info

> **NOTE:** Phones with fully randomized MACs and all ports blocked will still reach the hostname fallback. If the router assigns a model-name hostname (Fritz!Box, ASUS, TP-Link all do), the device will be identified. Devices that remain "unknown" after all three stages can be corrected via the Edit button on the agentless device row.

```
#### Stale Network Scan Cleanup (startup, not Celery)
```

Network scans run in a `threading.Thread(daemon=True)` – if Flask restarts mid-scan, the thread is killed but the `NetworkScan` DB record stays in `"running"` forever. On every `create_app()` call, a cleanup block runs **before** the API accepts requests:

```
cutoff = datetime.now(timezone.utc) - timedelta(minutes=5) stale = NetworkScan.query.filter(
NetworkScan.status == "running", NetworkScan.started_at < cutoff, ).all() for s in stale: s.status = "failed"
s.completed_at = datetime.now(timezone.utc) s.discovered_hosts = [{"error": "Scan interrupted (server
restarted)"}] db.session.commit()
```

This resolves all orphaned scans automatically on every API start. No manual DB intervention needed.

```
#### `ping_agentless_devices()`
- Loads all `Device` records where `is_agentless=True`
- Pings each device's `ip_address`
- Updates `is_online=True` and `last_seen=now()` on response
- Devices that have not responded for more than 10 minutes are marked `is_online=False`
- Changes committed in a single batch after all pings complete
```

```
### Starting Celery (Windows)
```

## Worker (must use `--pool=solo` on Windows — no `fork()`)

```
celery -A tasks.celery_app worker --pool=solo -l info
```

## Beat scheduler (separate process)

```
celery -A tasks.celery_app beat -l info
```

```
---
```

```
## 6. Agent Internals
```

```
### Cross-Platform Support
```

`collector.py` dispatches on `sys.platform` for all hardware/software/patch collection:`

| Platform | `sys.platform` | Hardware info                               | Patches                                | Software                                                            |
|----------|----------------|---------------------------------------------|----------------------------------------|---------------------------------------------------------------------|
| Windows  | `win32`        | WMI + winreg                                | WUA COM via PowerShell                 | Registry + winget                                                   |
| macOS    | `darwin`       | sysctl + system_profiler                    | platform.mac_ver()                     | softwareupdate -l + brew outdated   /Applications plist + brew list |
| Linux    | `linux`        | /proc/cpuinfo + dmidecode + /etc/os-release | apt-get/dnf/yum/pacman (auto-detected) | dpkg-query/rpm -qa/pacman -Q                                        |

The `platform` field in `hardware_info` is auto-detected (`windows`/`mac`/`linux`) instead of hardcoded. All Windows-specific code (winreg, WMI, DPAPI, winget) is guarded`

```
by `_PLATFORM == "win32"` checks.
```

```
### PyInstaller Binary Builder
```

```
`agent/build.py` produces a single-file executable with no Python dependency on the target machine:
```

```
pip install pyinstaller python build.py
```

## Output: dist/rmm\_agent.exe (Windows) or dist/rmm\_agent (Linux/macOS)

```
- `--onefile` mode, auto-detects `--noconsole` on Windows  
- Hidden imports per platform: `wmi`/`win32crypt`/`plistlib` included only when relevant  
- `config.ini` template copied next to binary after build  
- Deploy: copy `dist/` to target, edit `config.ini`, run binary
```

```
### Main Loop
```

startup ■■■ load agent\_state.json (device\_id, token) ■■■ if not registered: POST /api/agents/register → save state ■■■ prime CPU counter: psutil.cpu\_percent(interval=1) ■■■ loop every 60s: ■■■ collect\_metrics() (non-blocking cpu\_percent(interval=None)) ■■■ send\_heartbeat(metrics) ■■■ on 401: re-register ■■■ on connection error: exponential backoff 15s→300s ■■■ get\_tasks() ■■■ executor.run(task) for each queued task ■■■ if time – last\_patch > patch\_interval: ■■■ get\_pending\_patches() (platform-specific) ■■■ report\_patches(patches) ■■■ if time – last\_screenshot > screenshot\_interval (default 300s): ■■■ screenshot.capture() (PIL/scrot, None on headless) ■■■ heartbeat.send\_screenshot(jpeg\_bytes) ■■■ flush pending\_results.json (local queue)

```
### Task Execution
```

Agent receives tasks in format:

```
{ "type": "run_script", "run_id": "<uuid>", "script_type": "ps1", "script_content": "...", "timeout": 300 }
```

```
`executor.py` passes to `script_runner.py`:
```

```
- **ps1**: `powershell -NoProfile -NonInteractive -Command <content>` with  
  `CREATE_NO_WINDOW`  
- **bat**: `cmd /c <tempfile.bat>`  
- **py**: `python <tempfile.py>`  
- **sh**: `bash <tempfile.sh>` (Linux/macOS agents)
```

After execution, agent POSTs result to `/api/agents/<id>/task\_result`:

```
{ "run_id": "<uuid>", "exit_code": 0, "stdout": "...", "stderr": "", "status": "completed" }
```

```
### Patch Scanning
```

Uses PowerShell WUA COM object – requires no external package:

```
$session = New-Object -ComObject Microsoft.Update.Session $searcher =
$session.CreateUpdateSearcher() $result = $searcher.Search("IsInstalled=0 and Type='Software'")
```

Returns up to 500 patches as `[name, kb\_id, patch\_type]`. Reported via `PUT /api/agents/<id>/patches`. API deduplicates by `patch\_name` and creates `PatchRecord` rows with `status='pending'`.

### ### Software Collection

`get\_installed\_software()` merges two sources, deduplicated by lowercase name:

| Source           | Function                   | Notes                                                                      |
|------------------|----------------------------|----------------------------------------------------------------------------|
| Windows registry | `_get_registry_software()` | 3 hive paths, 20s hard deadline                                            |
| winget           | `_get_winget_software()`   | Skips non-ASCII lines (progress bars); finds separator line before parsing |

**\*\*Winget parser detail:\*\*** `winget list` prints Unicode block-char progress bars (■■■■, `ord > 127`) before the real data table. The parser skips these lines, finds the first all-dash separator line (length > 10), then parses only lines after it. The dashboard adds a `\_clean()` filter (strips U+2500-U+259F) as defence-in-depth before HTML rendering.

### ### Resilience Features

| Feature               | Implementation                                            |
|-----------------------|-----------------------------------------------------------|
| Non-blocking CPU      | `_cpu_percent(interval=None)` after startup prime         |
| Bounded process scan  | Cap at 200 processes, 3s timeout                          |
| Bounded registry scan | 20s hard deadline                                         |
| Winget Unicode parser | Skip `ord > 127` lines; locate separator line before data |
| Heartbeat backoff     | 15s → 30s → 60s ... → 300s cap on failure                 |
| 401 re-registration   | Main loop checks status code, triggers full re-register   |
| Local result queue    | `_pending_results.json`, cap 100, flushed each cycle      |
| Structured logging    | `_DeviceFilter` injects device_id into all log records    |

---

## ## 7. Built-in Scripts Reference

Built-in scripts are `Script` records with `is\_builtin=True`. The `description` field stores the lookup tag. Created/updated at API startup via `ensure\_builtin\_scripts()`.

| task_type       | Tag                         | Script name                     | What it does                                                                     |
|-----------------|-----------------------------|---------------------------------|----------------------------------------------------------------------------------|
| `clean_temp`    | `__builtin_clean_temp__`    | Built-in: Clean Temp Files      | Removes `%TEMP%`, `%TMP%`, `C:\Windows\Temp`                                     |
| `defrag`        | `__builtin_defrag__`        | Built-in: Defragment            | `Optimize-Volume -DriveLetter C -Defrag`                                         |
| `check_disk`    | `__builtin_check_disk__`    | Built-in: Check Disk            | `chkdsk C: /f` — schedules for next reboot if drive locked; exits 0 in all cases |
| `restore_point` | `__builtin_restore_point__` | Built-in: Create Restore Point  | `Checkpoint-Computer -Description "RMM"`                                         |
| `clear_browser` | `__builtin_clear_browser__` | Built-in: Clear Browser History |                                                                                  |

```
Removes Chrome/Edge/Firefox cache dirs |
| `reboot` | `__builtin_reboot__` | Built-in: Reboot | `Restart-Computer -Force` |
| `shutdown` | `__builtin_shutdown__` | Built-in: Shutdown | `Stop-Computer -Force` |
```

**\*\*Adding a new built-in script:\*\***

1. Add entry to `TASK\_TYPE\_TO\_TAG` dict in `api/utils/builtin\_scripts.py`
2. Add entry to `BUILTIN\_SCRIPTS` dict with `name`, `content`, `file\_type`
3. Restart the API — `ensure\_builtin\_scripts()` will create the DB record
4. Add the new `task\_type` to any dispatch logic that should use it

**\*\*Transient scripts\*\*** (not in the built-in registry):

- `\_\_deploy\_patches\_transient\_\_` — created dynamically by `deploy\_patches` Celery task per deployment run

---

## 8. Dashboard API Client

`dashboard/utils/api\_client.py` — `RMMClient` class

### Session Management

## One RMMClient per browser session, stored in `st.session_state`

```
def get_client(): if "_rmm_client" not in st.session_state: st.session_state["_rmm_client"] = RMMClient(...)
return st.session_state["_rmm_client"]
```

### Retry + 401 Refresh

```
def _request(self, method, path, **kwargs): for attempt in range(3): try: resp =
self.session.request(method, url, ...) if resp.status_code == 401 and attempt == 0: self._refresh_token()
continue return resp.json(), None except (ConnectionError, Timeout): time.sleep([0.5, 1.0, 2.0][attempt])
return None, "Service unavailable"
```

### Key Methods

| Method                                                 | HTTP | Path                               |
|--------------------------------------------------------|------|------------------------------------|
| `list_devices(page, per_page, **filters)`              | GET  | `/api/devices/`                    |
| `get_device(device_id)`                                | GET  | `/api/devices/<id>`                |
| `queue_device_task(device_id, task_type, timeout=300)` | POST | `/api/devices/<id>/queue_task`     |
| `reboot_device(device_id)`                             | POST | `/api/devices/<id>/reboot`         |
| `shutdown_device(device_id)`                           | POST | `/api/devices/<id>/shutdown`       |
| `deploy_patches(device_id, patch_ids)`                 | POST | `/api/devices/<id>/deploy_patches` |
| `list_tickets(**filters)`                              | GET  | `/api/tickets/`                    |
| `create_ticket(data)`                                  | POST | `/api/tickets/`                    |
| `update_ticket(ticket_id, data)`                       | PUT  | `/api/tickets/<id>`                |
| `add_comment(ticket_id, body, is_internal)`            | POST | `/api/tickets/<id>/comments`       |
| `list_alerts(**filters)`                               | GET  | `/api/alerts/`                     |

```
| `list_customers(per_page)` | GET | `/api/customers/` |
| `list_users()` | GET | `/api/users/` |
| `force_change_password(new_password)` | POST | `/api/auth/me/force-change-password` |
| `change_password(current_password, new_password)` | PUT | `/api/auth/me/password` |
| `mfa_setup()` | POST | `/api/auth/mfa/setup` |
| `mfa_enable(code)` | POST | `/api/auth/mfa/enable` |
| `mfa_disable(password)` | POST | `/api/auth/mfa/disable` |
| `mfa_login(mfa_token, code)` (static) | POST | `/api/auth/mfa/login` |
| `list_patches(**filters)` | GET | `/api/patches/` |
| `approve_patches(patch_ids)` | POST | `/api/patches/approve_bulk` |
| `list_profiles()` | GET | `/api/automation/profiles` |
| `run_profile(profile_id)` | POST | `/api/automation/profiles/<id>/run` |
| `create_report(data)` | POST | `/api/reports/` |
| `list_reports()` | GET | `/api/reports/` |
| `get_platform_counts()` | GET | `/api/devices/platform_counts` |
| `ping_check_device(device_id)` | POST | `/api/devices/<id>/ping_check` |
| `upsert_agentless_devices(hosts, customer_id=None)` | POST |
`/api/network/agentless_devices` |
| `trigger_network_scan(customer_id, scan_range)` | POST | `/api/network/scan` |
| `get_server_ips()` | GET | `/api/admin/server_ips` |
| `update_device(device_id, data)` | PUT | `/api/devices/<id>` |
| `delete_device(device_id)` | DELETE | `/api/devices/<id>` |
| `list_invoices(customer_id=None)` | GET | `/api/billing/invoices` |
| `get_invoice(invoice_id)` | GET | `/api/billing/invoices/<id>` |
| `generate_invoice(data)` | POST | `/api/billing/invoices` |
| `update_invoice_status(invoice_id, status)` | PUT | `/api/billing/invoices/<id>/status`
|
| `delete_invoice(invoice_id)` | DELETE | `/api/billing/invoices/<id>` |
| `get_invoice_pdf_bytes(invoice_id)` | GET | `/api/billing/invoices/<id>/pdf` – returns
`(bytes, error)`, bypasses JSON parsing |
| `send_invoice_email(invoice_id)` | POST | `/api/billing/invoices/<id>/send-email` |
| `get_org_settings()` | GET | `/api/admin/org-settings` |
| `update_org_settings(data)` | PUT | `/api/admin/org-settings` |
| `upload_org_logo(file_bytes, content_type)` | PUT | `/api/admin/org-settings/logo` –
uses `requests.put(files=)`, bypasses JSON headers |
| `delete_org_logo()` | DELETE | `/api/admin/org-settings/logo` |
```

### ### HTTPS Warning

`api\_client.py` logs a `WARNING` at import time if `API\_BASE\_URL` uses `http://` pointing to a non-localhost host. This warns that tokens and data will be transmitted unencrypted.

### ### Cache Strategy

@st.cache\_data(ttl=30) # fast-changing: devices, alerts @st.cache\_data(ttl=60) # medium: tickets, customers @st.cache\_data(ttl=120) # slow-changing: scripts, users, patch policies

---

## ## 9. Performance Optimizations

### ### Database

```

| Optimization | Location | Detail |
|-----|-----|-----|
| Composite indexes | Direct SQL (post-migration) | 7 new indexes:
`ix_device_metrics_device_collected_at`, `ix_devices_customer_online`,
`ix_alerts_device_status`, `ix_alerts_status_triggered_at`,
`ix_patch_records_device_status`, `ix_tickets_customer_status`,
`ix_script_runs_device_status` |
| N+1 alert evaluation fix | `tasks/alert_tasks.py` | `evaluate_all_rules()` now loads
all latest metrics per rule in 1 subquery. Was up to 5,000 queries/cycle (50 rules x 100
devices). |
| Metrics endpoint row cap | `routes/devices.py` | `GET /api/devices/<id>/metrics`
applies `.limit(5000)`. Was unbounded (10,080+ rows for 7-day window). |
| Batch latest metrics | `routes/devices.py:batch_latest_metrics()` | Subquery MAX(id)
per device – 1 query instead of N |
| Dashboard aggregation | `routes/dashboard.py` | 4 DB aggregations replace 8 COUNT
queries |
| Health map cap | `routes/dashboard.py` | `.limit(500)` prevents full-table load |
| N+1 agent tasks | `routes/agents.py` | Pre-fetch scripts by id dict before loop |
| DB connection pool | `config.py` | `pool_size=10, max_overflow=20, pool_pre_ping=True`
|

```

### ### Celery

```

| Optimization | Detail |
|-----|-----|
| `acks_late=True` | Task not acknowledged until completion – survives worker crash |
| `OperationalError` retry | DB connection drops retry with exponential backoff |
| Automation offloaded | Device-loop in `enqueue_profile_run` runs in Celery, not in
request thread |

```

### ### API Server

```

| Optimization | Detail |
|-----|-----|
| gzip compression | `flask-compress` gzips all JSON responses >500B. health_map:
5KB→1.2KB, activity_feed: 40KB→877B |
| Waitress threads | 16 threads (up from 8) – headroom for Celery beat contention |
| JWT decode cache | `api/utils/jwt_cache.py` – `TTLCache(512, 60s)` monkey-patches
`decode_jwt_from_request`. Repeated tokens skip HMAC: 60-86ms vs 135-177ms.
Process-local, no Redis needed. |
| Raw JSON cache | `cache_get_raw`/`cache_set_raw` stores pre-serialized JSON bytes in
Redis. `health_map` returns `Response(raw, mimetype="application/json")` – bypasses
double serialization. |
| Cache pre-warm | `create_app()` pre-populates `rmm:dash:summary:all` at startup so
first user hits Redis not DB. Cold hits: ~1300ms → ~100-260ms. |
| PgBouncer | Route `DATABASE_URL` through PgBouncer on port 6432 for persistent backend
connections. `pool_pre_ping` disabled (saves ~100ms per cold acquisition;
`pool_recycle=300` covers stale connections). |
| PostgreSQL work_mem | `ALTER ROLE rmm_app SET work_mem = '16MB'` – improves sort/hash
query performance. |

```

### ### Dashboard

```

| Optimization | Detail |
|-----|-----|
| `RMMClient` session reuse | One `requests.Session` per browser session – persistent HTTP connection |
| Retry backoff | `ConnectionError`/`Timeout` only – 0.5s/1.0s/2.0s |
| `dashboard/utils/cached_calls.py` | `st.cache_data` wrappers for all major read endpoints (`cached_summary` TTL 60s, `cached_health_map` TTL 30s, `cached_list_devices` TTL 30s, etc.). Call `st.cache_data.clear()` after mutations. |
| `st.fragment` auto-refresh | `01_Dashboard.py` body wrapped in `@st.fragment(run_every=30)` – no longer blocks server thread. `14_Terminal.py` output panel wrapped in `@st.fragment(run_every=2)`. |
| Parallel API calls | `04_Devices.py` and `10_Admin.py` use `ThreadPoolExecutor` to fire independent API calls in parallel. |
| `st.spinner` | All data loads wrapped – user sees progress feedback |
| Graceful degradation | `st.warning` instead of `st.stop()` – page stays interactive on partial failure |
| Token URL security | `require_auth()` strips `?tok=``?rtok=` from URL after first restore. Tokens no longer re-stamped on every page load – reduces browser history exposure. |
| Org settings session cache | `require_auth()` loads `GET /api/admin/org-settings` once per session into `st.session_state["org_settings"]`. All pages read `currency`/`timezone` from this dict – zero repeated API calls. Admin page clears `org_settings` from session state after Regional Settings save to force a fresh load. |

```

### ### Metrics History – 7-Day Fallback

`dashboard/pages/04\_Devices.py` – `\_render\_agent\_row()` metrics history block.

Default window: 24 hours (`GET /api/devices/<id>/metrics?hours=24`). If the API returns an empty list (agent was offline for >24h), the dashboard retries with `hours=168` (7 days). On fallback:

- An `st.info` banner shows: `"No data in last 24 h – showing last N readings (oldest ~Xh ago). Agent may be offline."`
- Chart title changes from `"24-hour usage history"` to `"7-day usage history (agent offline)"`.
- If the 7-day window is also empty, `st.warning("No metric history available.")` is shown.

The `collected\_at` timezone handling: `pd.to_datetime()` preserves tz-aware timestamps; the "hours ago" calculation converts to UTC via `.tz_convert("UTC")` before diffing against `pd.Timestamp.now(tz="UTC")`.

### ### Responsive Design – CSS Breakpoints

Responsive styles live entirely in `dashboard/utils/styles.py` (`_GLOBAL_CSS_RULES`) and are injected as a static CSS file via `inject_css()`. The login page adds its own mobile overrides through `_login_css()` in `dashboard/app.py`.

**\*\*Breakpoints:\*\***

```

| Breakpoint | Selector | Key rules |
|---|---|---|
| ≤1024px (tablet) | `@media (max-width: 1024px)` | Padding → 1.25rem, metric value →

```

```
1.6rem |
| ≤768px (mobile) | `@media (max-width: 768px)` | Columns stack (`flex-wrap: wrap`,
`flex: 1 1 100%`), 44px tap targets, 1rem input font, tabs scroll horizontally,
DataFrames scroll, sidebar capped at 82vw, login card full-width |
| ≤480px (small phone) | `@media (max-width: 480px)` | Padding → 0.4rem, heading/metric
further reduced |
```

**\*\*Column stacking – how it works:\*\***

Streamlit renders `st.columns()` as `[data-testid="stHorizontalBlock"]` with children  
`[data-testid="column"]`. On mobile:

```
[data-testid="stHorizontalBlock"] { flex-wrap: wrap !important; } [data-testid="column"] { width: 100%
!important; min-width: 100% !important; flex: 1 1 100% !important; }
```

All column layouts – stat cards, device rows, billing columns – stack vertically. Empty  
spacer columns (e.g. `\_`, `col`, `\_` in login) have no content so they take zero visible  
height.

**\*\*Touch targets (WCAG 2.5.5):\*\***

```
.stButton > button { min-height: 44px !important; } .stTextInput input { min-height: 44px !important;
font-size: 1rem !important; }
```

`font-size: 1rem` on inputs prevents iOS Safari from auto-zooming on focus.

**\*\*Tab overflow:\*\***

```
[data-testid="stTabs"] { overflow-x: auto !important; -webkit-overflow-scrolling: touch !important; }
[data-testid="stTabs"] [role="tablist"] { flex-wrap: nowrap !important; }
```

Tabs scroll horizontally on mobile instead of wrapping or clipping.

**\*\*Login page specifics (`\_login\_css()` in `app.py`):\*\***

Same column-stack rules are duplicated inside the login CSS block so they apply before  
`inject\_css()` is called. Login-specific: `min-height: 48px` on inputs/buttons (slightly  
larger for first-touch UX), `brand-title` shrinks to 1.5rem.

**\*\*Platform limitation:\*\*** Streamlit injects its own `
renders in an iframe context where the `window.innerWidth` seen by CSS is the iframe  
width, not the device width. On current Chrome/Safari (mobile), this equals the viewport  
width correctly. On some embedded or WebView contexts, breakpoints may not fire. No  
workaround exists within Streamlit's current architecture.

**\*\*White-label color swap compatibility:\*\*** `inject\_css()` does a string-replace of  
`#407E3C` → custom color before writing the CSS file. The responsive rules contain no  
color values, so the swap is unaffected.

**### Device Health Map – Clickable Cards**

Cards in `dashboard/pages/01\_Dashboard.py` use `st.button()` + `st.switch\_page()` for  
reliable Streamlit-native navigation. Plain `` links and `components.html()` with  
`window.parent.location.href` are intercepted/sandboxed by Streamlit's SPA router and do  
not work.

```

**Navigation flow:**
1. User clicks a health map card → `st.session_state["_nav_device"] = str(device_id)` → `st.switch_page("pages/04_Devices.py")`
2. `04_Devices.py` reads `st.session_state.pop("_nav_device", None)` on load, falls back to `st.query_params.get("device")` for direct URL access
3. If matched, pre-fills search box with the device hostname and shows an info banner

```

```

**CSS scoping:** Card button styles are injected inside the `else` branch of the health map render and use `section[data-testid="stMain"] button[kind="secondary"]` to target only main-area secondary buttons – sidebar buttons are excluded. Refresh button uses `type="primary"` to opt out of card styling entirely.

```

```

---

```

## ## 10. Extension Guide

### ### Adding a New Dashboard Page

1. Create `dashboard/pages/NN\_PageName.py`
2. Follow the standard page template:

```

import streamlit as st from utils.auth import require_auth from utils.nav import render_sidebar from
utils.styles import inject_css

```

```

st.set_page_config(page_title="Page Name — RMM", layout="wide") inject_css()

```

```

client = require_auth() render_sidebar()

```

```

st.markdown('<h1 style="margin:0">Page Name</h1>', unsafe_allow_html=True)

```

```

with st.spinner("Loading..."): data, err = client.some_method() if err: st.warning(f'Could not load data —
{err}') st.stop()

```

3. Add a nav link in `dashboard/utils/nav.py` under the appropriate section

### ### Adding a New API Route

1. Create `api/routes/mymodule.py`:

```

from flask import Blueprint, request, jsonify from flask_jwt_extended import jwt_required, get_jwt from
extensions import db

```

```

mymodule_bp = Blueprint("mymodule", __name__)

```

```

@mymodule_bp.route("/", methods=["GET"]) @jwt_required() def list_items(): ... return jsonify({"items":
[...]}), 200

```

2. Register in `api/app.py`:

```

from routes import mymodule app.register_blueprint(mymodule.mymodule_bp, url_prefix='/api/mymodule')

```

3. Add corresponding method to `dashboard/utils/api\_client.py`

### ### Adding a New Built-in Script

```
1. Edit `api/utils/builtin_scripts.py`:
```

## Add to TASK\_TYPE\_TO\_TAG:

```
"my_task": "__builtin_my_task__",
```

## Add to BUILTIN\_SCRIPTS:

```
"my_task": { "name": "Built-in: My Task", "content": "# PowerShell script here\nWrite-Output 'Done'",  
"file_type": "ps1", },
```

```
2. Restart the API – `ensure_builtin_scripts()` creates the DB record automatically  
3. Optionally add the `task_type` to `_dispatch_profile_tasks()` in  
`tasks/automation_tasks.py`  
4. Optionally add a UI button in the relevant dashboard page calling  
`client.queue_device_task(device_id, "my_task")`
```

```
### Adding WiFi/Agentless Devices
```

```
**For Windows/Linux/macOS laptops on the same WiFi/LAN:**
```

1. Find the server's LAN IP: Admin → System Info → Server IP Addresses, or check your router.
2. Copy the `agent/` folder to the target machine (USB, network share, or git clone).
3. On the target machine, install dependencies:

```
cd agent python -m venv venv venv\Scripts\activate # Windows pip install -r requirements.txt
```

4. Run the setup helper – patches `config.ini` automatically:

```
python setup_agent.py 192.168.x.x <org_token>
```

The org token is shown in Admin → System Info → Agent Enrollment Token.

5. Start the agent:

```
python rmm_agent.py
```

6. The device appears in Devices → Windows/Linux/macOS tab within 60 seconds.

```
**For phones, tablets, and IoT devices (no agent possible):**
```

1. Go to Network Discovery in the dashboard.
2. Enter the subnet in CIDR notation (e.g. `192.168.1.0/24`) and optionally assign a customer.
3. Click **Scan Network** – the scan takes 15–30 seconds via concurrent ICMP ping.
4. Discovered devices appear with IP, MAC, vendor (from OUI lookup), and detected platform.
  - iOS devices are identified by Apple OUI prefixes, or by port 62078 being open (port probe fallback).
  - Android devices are identified by Samsung/Google/OnePlus/etc. OUI prefixes, or by port 5555 being open.
  - When OUI lookup returns unknown, `\_probe\_platform(ip)` tries 6 TCP ports to detect the

OS.

- Reverse DNS (``_get_hostname``) populates a friendly hostname (e.g. ``Printer.local``) when available.

5. Click **Save All to Devices** to persist them to the database.

6. Devices appear in Devices → Android/iOS/Agentless tabs.

7. Celery beat pings them every 5 minutes – online/offline status updates within 10 minutes of disconnect.

**Identifying devices that evade automated detection (randomized MAC + closed ports):**

Modern phones randomize their MAC address per network, defeating OUI lookup. They also typically block all probed ports. These devices will remain platform "unknown" after a scan. Use the **Edit** button on the agentless device row in the Devices page to manually set hostname, platform, and device\_type. This calls ``PUT /api/devices/<id>`` via ``client.update_device(device_id, data)``.

### Adding or Changing a Supported Currency or Timezone

All supported currencies and timezones are defined as Python lists in ``dashboard/utils/formatters.py``:

```
SUPPORTED_CURRENCIES = ["USD", "EUR", "GBP", "CHF", "CAD", "AUD", "JPY", "SEK", "NOK", "DKK"]
```

```
CURRENCY_SYMBOLS = { "USD": "$", "EUR": "€", "GBP": "£", "CHF": "Fr", "CAD": "CA$", "AUD": "A$", "JPY": "¥", "SEK": "kr", "NOK": "kr", "DKK": "kr", }
```

```
SUPPORTED_TIMEZONES = [ "UTC", "Europe/Berlin", "Europe/London", ... ]
```

To add a new currency: append the ISO code to ``SUPPORTED_CURRENCIES`` and add its symbol to ``CURRENCY_SYMBOLS``. No DB migration needed – currency/timezone values are stored as plain strings.

To add a new timezone: append the IANA name to ``SUPPORTED_TIMEZONES``. Verify Python's ``zoneinfo`` module recognizes it: ``from zoneinfo import ZoneInfo; ZoneInfo("Your/Zone")`` (raises ``ZoneInfoNotFoundError`` if invalid).

**``fmt_currency(amount, currency)``** – formats a numeric amount with the correct symbol:

```
fmt_currency(49.99, "EUR") # → "€49.99" fmt_currency(49.99, "JPY") # → "¥49.99"
```

**``fmt_datetime_tz(iso_str, tz_name)``** – converts a UTC ISO string to local time for display:

```
fmt_datetime_tz("2026-06-16T10:00:00Z", "Europe/Berlin") # → "2026-06-16 12:00"
```

```
fmt_datetime_tz("2026-06-16T10:00:00Z", "America/New_York") # → "2026-06-16 06:00"
```

Falls back to ``fmt_datetime(iso_str)`` (UTC, no conversion) if ``zoneinfo`` raises or the ``tz_name`` is invalid. This prevents crashes when an unsupported timezone is stored in the DB.

**Consuming regional settings in a new dashboard page:**

```
_org = st.session_state.get("_org_settings", {}) _currency = _org.get("currency", "USD") _tz =
_org.get("timezone", "UTC")
```

## Then use:

```
fmt_currency(invoice["total"], _currency) fmt_datetime_tz(invoice["created_at"], _tz)
```

`\_org\_settings` is loaded once per session by `require\_auth()` and is always available by the time any page body runs.

### Adding a New Report Template

1. Add to `tasks/report\_tasks.py`:

```
def _my_template(report, db_session): rows = db_session.query(MyModel).all() return [{"col1": r.field1,
"col2": r.field2} for r in rows]
```

2. Register in the dispatcher dict inside `generate\_report()`:

```
COLLECTORS = { ... "my_template": _my_template, }
```

3. Add option to the template dropdown in `dashboard/pages/08\_Reports.py`

```
---
```

## 11. Security Model

### Network Boundaries

| Surface            | Exposure           | Notes                                       |
|--------------------|--------------------|---------------------------------------------|
| Streamlit `:8501`  | LAN/localhost only | No direct user data – proxies to API        |
| Flask API `:5000`  | LAN/localhost only | All auth enforced via JWT                   |
| PostgreSQL `:5432` | localhost only     | `rmm_app` user has no superuser             |
| Redis `:6379`      | localhost only     | No password in dev (add one for production) |

### Authentication Layers

1. **Dashboard users** – JWT (HS256, `JWT\_SECRET\_KEY`), 900s access / 7d refresh
2. **Agents** – `X-Agent-Token` header, token stored as SHA256 hash in `agent\_tokens` table
3. **API startup** – `ORG\_REGISTRATION\_TOKEN` required for agent registration
4. **Superadmin** – seeded at every API startup via `ensure\_superadmin()` in `api/utils/superadmin.py`; bypasses all `\_require\_role()` checks; cannot be deleted or modified via API; password reset via `api/reset\_superadmin.py` CLI

### Known Risks & Mitigations

| Risk            | Mitigation                                                                                                                                                                      |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JWT token theft | Short expiry (900s). HTTPS in production. Tokens written to URL once at login, then stripped by `require_auth()` after first restore – no longer re-stamped on every page load. |

| Script injection | Scripts stored as plain text – reviewed before running. Agent now uses ``-ExecutionPolicy RemoteSigned`` instead of ``Bypass`` – locally created scripts run, remote scripts must be signed. |

| Error response leakage | 400/422 handlers no longer return ``str(e)`` to client. Detail logged server-side only; generic message returned to caller. |

| CORS wildcard | ``CORS_ORIGINS`` env var (comma-separated) replaces ``origins=""``. Default: ``http://localhost:8501``. Set to dashboard URL(s) in production. |

| Privilege escalation | Role enforced at API level via ``get_jwt()`` claims check on every mutation endpoint |

| Celery task injection | Tasks only dispatched by authenticated API endpoints or beat schedule |

| DB credentials in env | ``.env`` in ``.gitignore``. Never committed. |

| Report file exposure | ``api/reports/`` not served via HTTP. Download reads bytes directly in dashboard. |

| SMTP credential exposure | ``SMTP_PASS`` read from env only. Never logged. |

| Superadmin lockout | Auto-seeded at every API startup – account always exists. Emergency password reset: ``cd api ; python reset_superuser.py <new_password>`` |

#### ### Production Checklist

- [ ] Set ``FLASK_DEBUG=0``
- [ ] Use HTTPS (nginx reverse proxy with Let's Encrypt)
- [ ] Add Redis password: ``requirepass <password>`` in Memurai config
- [ ] Rotate ``ORG_REGISTRATION_TOKEN`` after all agents registered
- [ ] Change default admin password from ``Admin1234!``
- [ ] Set up automated PostgreSQL backups
- [ ] Move ``api/reports/`` to a dedicated volume outside the project root
- [ ] Run agent as a dedicated low-privilege Windows service account (not SYSTEM)
- [ ] Enable MFA for all admin accounts (full TOTP implementation: setup, enable, login, disable via Profile page)
- [ ] Set ``CORS_ORIGINS`` to dashboard URL (replaces wildcard ``origins=""``)
- [ ] Set ``SUPERADMIN_PASSWORD`` in ``.env`` (now required -- API will not start without it)

#### ### Dashboard Session Storage (fixed 2026-09-22)

The 2026-09-22 security audit flagged that dashboard tokens rode continuously in the URL (``?tok=/'&rtok=``) to survive Streamlit page reloads – landing in browser history and any server/proxy access log that records query strings. This is now fixed properly rather than just mitigated:

- New ``dashboard/utils/session_store.py`` – the real access/refresh tokens live server-side in Redis (``dash:session:<id>``, TTL matching ``JWT_REFRESH_TOKEN_EXPIRES``), keyed by a random opaque session id (``secrets.token_urlsafe(32)``). Only that id goes in the URL (``?sid=``) – a leaked/logged URL is now useless on its own.
- ``dashboard/utils/auth.py::establish_session()`` is the single entry point every login flow (password, MFA) calls to create a session and stamp ``?sid=``. ``require_auth()`` keeps it in sync on reload; ``api_client.py``'s silent token-refresh path updates the stored tokens in place so a rotated access token survives the next reload too.
- Degrades gracefully if Redis is unreachable: falls back to the old raw-token-in-URL scheme (logged as a warning) rather than breaking login. Legacy ``?tok=/'&rtok=`` links (already-open tabs from before this change) are still accepted and migrated to a session on next load.
- Requires the dashboard process to reach the same Redis instance as the API (``REDIS_URL``)

env var, same convention as `api/utils/cache.py`) – new `redis==5.2.1` dependency in `dashboard/requirements.txt`.

### Known Residual Risk (audited 2026-09-22, not fixed)

- **\*\*React frontend stores the JWT in `localStorage`\*\***  
(`frontend/src/contexts/AuthContext.tsx`, `frontend/src/api/client.ts`) – exfiltrable by any successful XSS on that page. **\*\*Not fixed.\*\*** The correct fix is httpOnly-cookie-issued JWTs with CSRF token handling, which changes how `api/routes/auth.py` issues tokens for every client (Streamlit dashboard included) – a cross-cutting auth redesign, tracked here as a named follow-up rather than attempted piecemeal.

---

## ## 12. Environment Variables Reference

| Variable                                            | Required | Default                    | Description                                                                                          |
|-----------------------------------------------------|----------|----------------------------|------------------------------------------------------------------------------------------------------|
| ----- ----- ----- -----                             |          |                            |                                                                                                      |
| `DATABASE_URL`                                      | ✓        | –                          | PostgreSQL connection string                                                                         |
| `SECRET_KEY`                                        | ✓        | –                          | Flask secret key (32-byte hex)                                                                       |
| `JWT_SECRET_KEY`                                    | ✓        | –                          | JWT signing key (32-byte hex)                                                                        |
| `ORG_REGISTRATION_TOKEN`                            | ✓        | –                          | Agent registration shared secret                                                                     |
| `JWT_ACCESS_TOKEN_EXPIRES`                          | –        | `900`                      | Access token TTL in seconds                                                                          |
| `JWT_REFRESH_TOKEN_EXPIRES`                         | –        | `604800`                   | Refresh token TTL in seconds                                                                         |
| `REDIS_URL`                                         | –        | `redis://localhost:6379/0` | Redis connection URL                                                                                 |
| `CELERY_BROKER_URL`                                 | –        | `redis://localhost:6379/0` | Celery broker                                                                                        |
| `CELERY_RESULT_BACKEND`                             | –        | `redis://localhost:6379/1` | Celery result backend                                                                                |
| `API_BASE_URL`                                      | –        | `http://localhost:5000`    | Used by dashboard to reach API                                                                       |
| `SMTP_HOST`                                         | –        | –                          | SMTP server (omit to disable email alerts)                                                           |
| `SMTP_PORT`                                         | –        | `587`                      | SMTP port                                                                                            |
| `SMTP_USER`                                         | –        | –                          | SMTP login username                                                                                  |
| `SMTP_PASS`                                         | –        | –                          | SMTP login password (use app-specific password)                                                      |
| `SMTP_FROM`                                         | –        | `SMTP_USER`                | From address for alert emails                                                                        |
| `SUPERADMIN_EMAIL`                                  | –        | `superadmin@rmm.local`     | Email for the auto-seeded superadmin account                                                         |
| `SUPERADMIN_PASSWORD`                               | ✓        | –                          | <b>**Now required at startup.**</b> Min 10 chars. API raises `RuntimeError` if missing or too short. |
| `CORS_ORIGINS`                                      | –        | `http://localhost:8501`    | Comma-separated allowed CORS origins. Set to dashboard URL(s) in production.                         |
| `DB_PASSWORD`                                       | –        | `changeme`                 | PostgreSQL password used in `docker-compose.yml` only.                                               |
| `FLASK_ENV`                                         | –        | `production`               | Set to `development` for debug mode                                                                  |
| `FLASK_DEBUG`                                       | –        | `0`                        | Set to `1` only in development                                                                       |
| `SENTRY_DSN`                                        | –        | –                          | Sentry DSN. Omit to disable. No-op when unset.                                                       |
| `traces_sample_rate=0.05`, `send_default_pii=False` |          |                            |                                                                                                      |
| `DASHBOARD_URL`                                     | –        | `http://localhost:3000`    | React frontend URL – used in CORS and Stripe redirect URLs                                           |
| `MQTT_HOST`                                         | –        | –                          | MQTT broker hostname. Omit to disable MQTT ingestion.                                                |
| `MQTT_PORT`                                         | –        | `1883`                     | MQTT broker port                                                                                     |
| `MQTT_USERNAME`                                     | –        | –                          | MQTT broker credentials (optional)                                                                   |
| `MQTT_PASSWORD`                                     | –        | –                          | MQTT broker credentials (optional)                                                                   |
| `MQTT_TOPIC_PREFIX`                                 | –        | `rmm`                      | Topic pattern: `{prefix}/{device_id}/sensors/{type}`                                                 |
| `SNMP_TIMEOUT`                                      | –        | `3`                        | SNMP GET request timeout in seconds                                                                  |

```
| `STRIPE_SECRET_KEY` | - | - | Stripe secret key for payment processing |
| `STRIPE_WEBHOOK_SECRET` | - | - | Stripe webhook signing secret |
| `LATEST_AGENT_VERSION` | - | `1.0.0` | Current agent version for auto-update checks |
| `AGENT_UPDATE_PATH` | - | - | URL path to download updated agent binary |
```

Generate secrets:

```
python -c "import secrets; print(secrets.token_hex(32))" # SECRET_KEY, JWT_SECRET_KEY
python -c "import secrets; print(secrets.token_hex(24))" # ORG_REGISTRATION_TOKEN
```

---

## 13. Docker Deployment

### docker-compose.yml (project root)

6 services:

```
| Service | Image | Command |
|-----|-----|-----|
| `db` | `postgres:16-alpine` | Postgres with health check |
| `redis` | `redis:7-alpine` | Redis with health check |
| `api` | Built from `api/Dockerfile` | `flask db upgrade && python app.py` |
| `celery_worker` | Same as `api` | `celery -A tasks.celery_app worker --pool=solo -l info` |
| `celery_beat` | Same as `api` | `celery -A tasks.celery_app beat -l info` |
| `dashboard` | Built from `dashboard/Dockerfile` | `streamlit run app.py` |
```

### Dockerfiles

```
- `api/Dockerfile` - `python:3.11-slim` + `libpq-dev gcc` (for psycopg2 compile)
- `dashboard/Dockerfile` - `python:3.11-slim`
```

### Usage

## Start all services

---

`docker-compose up -d`

## View API logs

---

`docker-compose logs -f api`

## Stop (preserves data volume)

`docker-compose down`

## Stop + delete all data

`docker-compose down -v`

## Rebuild after code changes

`docker-compose build api dashboard && docker-compose up -d`

### Environment note

When using Docker Compose, `DATABASE_URL` must use `@db:5432` (service name), not `@localhost:5432`.  
`REDIS_URL` / `CELERY_BROKER_URL` must use `redis://redis:6379/...`.  
`SUPERADMIN_PASSWORD` is required – API container will exit immediately if missing.

### Flask Reloader Stability (`use_reloader=False`)

`app.run()` sets `use_reloader=False` permanently:

```
app.run( host=os.getenv("API_HOST", "0.0.0.0"), port=int(os.getenv("API_PORT", 5000)),
debug=os.getenv("FLASK_DEBUG", "1") == "1", use_reloader=False, )
```

**Why:** Flask's default `watchdog` reloader monitors all files reachable via `sys.path`, including Redis/Memurai's site-packages (`resp2.py`, `resp3.py`, etc.). Memurai writes to these files during normal operation, causing Flask to restart every ~30 seconds – killing in-flight network scan threads, resetting all connections, and returning 500/connection-reset errors to the dashboard.

`use_reloader=False` eliminates these restarts. Code changes require a manual API restart during development. In production, this is the expected behavior regardless.

### Security hardening applied at startup (`api/app.py`)

`_validate_env()` runs at import time and raises `RuntimeError` if:

- `SECRET_KEY` or `JWT_SECRET_KEY` < 32 characters
- `SUPERADMIN_PASSWORD` unset or < 10 characters
- `ORG_REGISTRATION_TOKEN` is the placeholder value `"replace-with-a-unique-org-token"`

Additional hardening:

- `X-Request-ID` header: `before_request` generates UUID if not present; echoed on all responses.
- Dev mode warning: API logs WARNING on startup if `FLASK_ENV != "production"`.
- CORS restricted via `CORS_ORIGINS` env var (replaces `origins=""`).

---

```
## 14. AI Assistant
```

```
### Architecture
```

[Streamlit page] ■ `render_ai_assistant(page_name, context)` ▼ [dashboard/utis/ai\_assistant.py] ■ POST /api/assistant/chat (JWT Bearer) ▼ [api/routes/assistant.py] ■ `_build_system_prompt(role, page, context)` ■ `anthropic.Anthropic(api_key=...).messages.create(...)` ▼ [Anthropic API — claude-haiku-4-5-20251001] ■ reply text + suggested\_actions list ▼ [Streamlit sidebar chat panel]

```
### New files
```

```
| File | Role |
|-----|-----|
| `api/routes/assistant.py` | Flask blueprint, chat endpoint, system prompt builder |
| `dashboard/utis/ai_assistant.py` | Streamlit sidebar widget |
```

```
### `POST /api/assistant/chat`
```

```
**Auth:** JWT Bearer (same as all other API endpoints)
**Rate limit:** 30 requests per minute per IP (Flask-Limiter)

**Request body:**
```

```
{ "message": "How do I deploy patches?", "page": "OS Patches", "context": {"pending_patches": "12",
"selected_device": "DESKTOP-01"}, "history": [{"role": "user", "content": "..."}, {"role": "assistant", "content":
"..."}]}
```

```
**Response:**
```

```
{ "reply": "To deploy patches: 1. Select a device...", "suggested_actions": ["Deploy OS patches", "Check
patch status", "What devices need updates?"] }
```

```
**Constraints:**
- `message` max 1000 chars (truncated server-side)
- `history` capped at last 10 turns
- `context` keys/values sanitised: key max 60 chars, value max 300 chars
- `page` max 60 chars
```

```
**Error responses:**
```

```
| Status | Condition |
|-----|-----|
| 400 | `message` empty |
| 429 | Rate limit exceeded |
| 503 | `AI_ASSISTANT_ENABLED=false`, or `ANTHROPIC_API_KEY` missing/invalid, or
Anthropic API down |
```

```
### System prompt structure
```

```
`_build_system_prompt(role, page, context)` assembles:
```

You are the AI Assistant embedded in Remote Management System (RMS)...

USER ROLE: {role} ROLE PERMISSIONS: {role\_desc}

CURRENT PAGE: {page} PAGE PURPOSE: {page\_desc}

LIVE CONTEXT:

- {key}: {value}

...

RESPONSE RULES:

1. Concise — 2-4 sentences for simple answers; numbered steps for how-to.
2. Use exact UI element names (button labels, tab names, headings).
3. Only describe features that exist. Never invent pages, buttons, or endpoints.
4. Respect role — if feature requires higher role, say so.
5. For viewers: never suggest create/edit/delete/run actions.
6. Off-topic: "I can only help with navigating this system."
7. Plain language — user may be new to RMM tools.
8. Multi-step tasks: number the steps.
9. If context shows a problem, mention the most urgent action first.

```
### Role extraction from JWT
```

```
from flask_jwt_extended import jwt_required, get_jwt
```

```
@assistant_bp.route("/chat", methods=["POST"]) @jwt_required() def chat(): role = get_jwt().get("role",  
"viewer") # role is in additional_claims ...
```

```
> **Critical:** `get_jwt_identity()` returns the user ID (int), not the role. Role lives  
in `additional_claims`. Always use `get_jwt()`.
```

```
### Dashboard widget
```

```
`render_ai_assistant(page_name, context)` in `dashboard/utils/ai_assistant.py`:
```

```
from utils.api_client import RMMClient
```

```
def render_ai_assistant(page_name="Overview", context=None): token =  
st.session_state.get("access_token") if not token: return # not logged in — silently skip client =  
RMMClient( access_token=token, refresh_token=st.session_state.get("refresh_token", ""), )
```

## ... initialise session state, render toggle, chat history, form, suggestions

```
> **Critical:** Do NOT use `st.session_state.get("_rmm_client")` – that key is never set.
`require_auth()` returns the client but does not store it. Build `RMMClient` directly
from `access_token`.

**Session state keys:**

| Key | Type | Purpose |
|-----|-----|-----|
| `_ai_history` | `list[dict]` | Chat turns `{role, content}` – capped display at 14 |
| `_ai_open` | `bool` | Panel open/closed |
| `_ai_seen_onboard` | `bool` | Whether welcome message has been shown this session |
| `_ai_suggested` | `list[str]` | Quick action button labels from last API response |

**Page key:** widget keys are scoped to `page_name.lower().replace(" ", "_").replace("/",
"_")` to prevent duplicate-key errors across pages.

### Adding the widget to a page
```

```
from utils.ai_assistant import render_ai_assistant
```

## At the bottom of the page (after all page content):

```
render_ai_assistant("Page Name")
```

## With live context (data-heavy pages):

```
render_ai_assistant("Overview", { "total_devices": total, "online_devices": online, "open_alerts": alerts, })
```

```
The widget renders inside `with st.sidebar:` – no layout columns needed.

### Environment variables

| Variable | Required | Default | Description |
|-----|-----|-----|-----|
| `ANTHROPIC_API_KEY` | Yes | – | Anthropic API key (`sk-ant-api03-...`) |
| `AI_ASSISTANT_MODEL` | No | `claude-haiku-4-5-20251001` | Model ID |
| `AI_ASSISTANT_ENABLED` | No | `true` | Set `false` to disable without removing code |

### Dependency
```

## api/requirements.txt

anthropic>=0.40.0

```
Install: `pip install anthropic>=0.40.0`
```

### ### Security properties

- API key stored server-side only (`api/.env`) – never sent to dashboard or browser
- JWT required on `/api/assistant/chat` – unauthenticated requests return 401
- Rate limiting prevents API cost abuse (30 req/min per IP via Flask-Limiter)
- Conversation history stored in Streamlit session state only – cleared on logout/refresh, never persisted to DB
- Context values sanitised before inclusion in system prompt

### ### Extension points

**\*\*Add a new page description:\*\*** Add an entry to `\_PAGE\_INFO` in `api/routes/assistant.py`. Key must match the `page\_name` string passed from the dashboard page.

**\*\*Add quick actions for a page:\*\*** Add an entry to `\_PAGE\_ACTIONS` with the same key.

**\*\*Add role-aware behaviour:\*\*** Update `\_ROLE\_CAPABILITIES` dict.

**\*\*"Do it for me" actions (future):\*\*** The widget already holds a `RMMClient` instance. Wire suggested-action buttons to call `client.create\_ticket(...)` etc. instead of just sending a chat message.

**\*\*Voice input (future):\*\*** Add `st.audio\_input()` → transcribe → feed transcript to `\_send\_message()`. No backend changes needed.

- 400/422 error handlers: detail logged server-side only, generic message returned to client.

---

## ## 16. Commercial Audit – Phase F

### ### F.1 Agent Multi-Customer Routing

`api/schemas/agents.py` – `customer\_id` = `fields.String(load\_default="", validate=validate.Length(max=36))`

`api/routes/agents.py` registration – if `customer\_id` in payload and valid UUID: assigns agent to that customer. Otherwise falls back to `Customer.query.filter\_by(is\_active=True).first()` (backward compatible).

`agent/config.ini` – `customer\_id` = under `[api]`. Set via `setup\_agent.py <server\_ip> <org\_token> <customer\_id>`.

### ### F.2 DPAPI Agent Token Encryption

`agent/rmm\_agent.py` – `\_protect\_token()` / `\_unprotect\_token()`. Windows: encrypts with

``win32crypt.CryptProtectData``, stores as ``DPAPI:<base64>``. Non-Windows or error: stores plaintext. Existing plaintext tokens auto-migrate on first write. Requires ``pywin32`` on Windows.

### ### F.3 OpenAPI 3.0 / Swagger

``GET /api/docs`` – SwaggerUI (CDN, brand topbar). ``GET /api/openapi.json`` – raw spec (47 paths, 16 tags, BearerAuth). No new pip deps.

### ### F.4 Redis Cache (``api/utils/cache.py``)

``cache_get(key)`` / ``cache_set(key, value, ttl)`` / ``cache_delete(key)``. JSON-serialized Redis values with TTL. Silent no-op on Redis unavailability.

Applied: dashboard summary (30s, per-tenant key), platform\_counts (60s).

### ### F.5 Alert Thundering Herd Fix

Redis ``SET NX EX 55`` lock in ``evaluate_all_rules``. Lock key: ``rmm:lock:evaluate_all_rules``. If lock not acquired, task exits. Released in ``finally``.

### ### F.6 Webhook Dispatch (``api/utils/webhook.py``)

``dispatch_alert_webhooks(channels, rule_name, device_hostname, message, severity)``.

Formats: Slack Block Kit (color-coded), MS Teams MessageCard, generic JSON POST. 5s timeout. Never raises.

### ### Terminal Improvements

| Fix | Detail |  
|-----|-----|  
| Output polling | ``TerminalPage.tsx`` polls ``GET /terminal/sessions/<id>/output?after=<last_id>`` every 2s instead of reading from the POST /commands response (which only returns a pending status object, not output). |  
| Pending-command indicator | ``GET /terminal/sessions/<id>/output`` now returns ``pending_commands`` count. Dashboard shows amber blinking indicator when commands are queued but agent has not yet picked them up. |  
| Alert storm fix | ``evaluate_all_rules`` uses separate ``open_alert_map`` (no time filter) and ``resolved_alert_map`` (cooldown after recovery) – prevents indefinite re-fire of open alerts that aged past ``cooldown_minutes``. |  
| Token rotation | ``terminal_worker.update_token()`` called from ``rmm_agent`` on token rotation – terminal worker no longer continues with a revoked token. |  
| Session seq reset | Auto-close fragment resets ``_term_output_seq`` to 0 on session close – prevents stale buffer on next connect. |

### ### F.7 Device Ownership (Client Role)

- ``POST /api/terminal/sessions`` – returns 404 if ``device.customer_id != claims["customer_id"]``
- ``POST /api/scripts/<id>/run`` – filters ``device_ids`` to customer-scoped query; unauthorized IDs skipped

### ### F.8 Script Run Audit

```
`run_script()` emits `AuditLog(action="script_run", resource_type="script",
payload={script_name, device_count, device_ids})` after commit.
```

### ### F.9 GDPR Endpoints

```
| Endpoint | Behaviour |
|-----|-----|
| `GET /api/admin/users/<id>/gdpr-export` | Returns `{user, audit_log[1000],
ticket_comments[1000]}` |
| `DELETE /api/admin/users/<id>/gdpr-delete` | Anonymizes PII, scrubs IPs from audit log,
nulls comment author email. Irreversible. |
```

Both admin-only. Both emit `AuditLog` entry.

### ### F.10 Billing Automation

`Customer` gains `billing\_day`, `per\_device\_rate`, `tax\_rate` (migration `j1k2l3m4n5o6`).

`tasks/billing\_tasks.py` `generate\_recurring\_invoices` (beat 86400s): generates draft invoice for prior month for customers where `billing\_day == today` and `per\_device\_rate > 0`. Idempotent on `(customer\_id, period\_start)`.

### ### F.11 Configurable SLA Policies

`SLAPolicy` model: `(customer\_id nullable, priority)` unique. Seeded with 4 global defaults.

Routes at `/api/sla-policies/`. `tickets.py` `\_sla\_resolution\_hours()` lookup: customer-specific → global → hardcoded.

### ### F.12 Database Backup (`tasks/backup\_tasks.py`)

`backup\_database` (beat 86400s): pg\_dump → gzip → `BACKUP\_DIR`. Prunes files older than `BACKUP\_RETAIN\_DAYS` days. Retry on timeout (3600s) or error (1800s).

Env: `BACKUP\_DIR` (default `../backups`), `BACKUP\_RETAIN\_DAYS` (default 7).

Restore: `gunzip -c backups/rmddb\_YYYYMMDD\_HHMMSS.sql.gz | psql -U rmm\_app rmddb`

### ### F.13 Migration Chain (complete)

d746c05420bd → 2c8f1d3e9a4b → 93baa3927b0c → f3e2d1c0b9a8 → a1b2c3d4e5f6 → e5d4c3b2a1f0  
→ b5c4d3e2f1a0 → c1d2e3f4a5b6 → d2e3f4a5b6c7 → e6f7a8b9c0d1 → f1a2b3c4d5e6 → a2b3c4d5e6f7  
→ b3c4d5e6f7a8 → c4d5e6f7a8b9 → d5e6f7a8b9c0 → f7a8b9c0d1e2 → h9i0j1k2l3m4 → i0j1k2l3m4n5  
→ j1k2l3m4n5o6 → k2l3m4n5o6p7 → l3m4n5o6p7q8 (IoT sensor\_readings table)

---

## ## 15. React Frontend

### ### Stack

| Package        | Version | Role                    |
|----------------|---------|-------------------------|
| React          | 19      | UI framework            |
| Vite           | 8       | Dev server + build tool |
| TypeScript     | 6       | Type safety             |
| TanStack Query | 5       | Server state + caching  |
| React Router   | 7       | Client-side routing     |
| Tailwind CSS   | 3       | Utility-first styling   |
| axios          | 1.x     | HTTP client             |
| lucide-react   | 1.x     | Icon library            |

### ### Pages

All 19 pages match Streamlit equivalents: `DashboardPage`, `TicketsPage`, `CustomersPage`, `DevicesPage`, `AlertsPage`, `AdminPage`, `AutomationPage`, `BillingPage`, `ClientPortalPage`, `DiskManagementPage`, `MaintenancePage`, `NetworkPage`, `OSPatchesPage`, `ProfilePage`, `ReportsPage`, `ScriptsPage`, `SoftwarePatchesPage`, `TerminalPage` (with output polling fix applied), `auth/` (login flow).

### ### Running

cd frontend npm install npm run dev # http://localhost:3000 npm run build # production build → frontend/dist/ npm run lint # oxlint

### ### CORS configuration

Set `CORS\_ORIGINS=http://localhost:3000` in `api/.env`. For production, include both the Streamlit and React URLs:

CORS\_ORIGINS=http://localhost:8501,https://dashboard.example.com

### ### Auth flow

Same JWT Bearer token flow as Streamlit. Tokens stored in `localStorage` (React) vs `st.session\_state` + URL params (Streamlit). The React frontend uses `AuthContext` to hold `access\_token` and `refresh\_token`; axios interceptors handle 401 → refresh → retry automatically.

---

## ## 16. Screenshot Capture Pipeline

### ### Agent side (`agent/screenshot.py`)

`capture()` returns JPEG bytes or `None`:

- **Windows/macOS:** `PIL.ImageGrab.grab()` (Pillow)
- **Linux:** `scrot` subprocess → `gnome-screenshot` → PIL fallback
- Resizes to max 1920px wide, JPEG quality 72 (~100-400 KB per capture)
- Returns `None` silently on headless/no-display – non-fatal

```
Periodic sync in `rmm_agent.py`:
```

```
screenshot_interval = config.getint("agent", "screenshot_interval", fallback=300)
```

All errors caught at DEBUG level – never breaks the heartbeat loop.

### API endpoints

| Method | Path                                                    | Auth          | Description                                                                                                                                                     |
|--------|---------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POST   | <code>/api/agents/&lt;device_id&gt;/screenshot`</code>  | X-Agent-Token | Accepts <code>image/jpeg`</code> or <code>image/png`</code> , 5 MB cap. Saves to <code>api/screenshots/&lt;device_id&gt;.jpg`</code> . Replaces on each upload. |
| GET    | <code>/api/devices/&lt;device_id&gt;/screenshot`</code> | JWT           | Returns latest screenshot via <code>send_file`</code> . 404 JSON error if no screenshot captured yet.                                                           |

Screenshots are stored in `api/screenshots/`` (gitignored, `.gitkeep`` present). One file per device – only the latest capture is retained. The directory is not served via HTTP; the API streams the file through `send_file``.

### Dashboard

The Devices page (agent device rows) shows the latest screenshot inline when available. The React `DevicesPage`` fetches `GET /api/devices/<id>/screenshot`` and renders it as an `<img>`` tag.

---

## 17. IoT / MQTT / SNMP

### IoT Sensor Agent (`agent/iot_agent.py``)

Lightweight sensor collector for Raspberry Pi / Linux SBC. Reuses `agent/config.ini`` and `heartbeat.py`` APIClient for auth.

Sensor backends (all optional – skipped if library not installed):

| Backend                        | Library                                     | Sensors                       |
|--------------------------------|---------------------------------------------|-------------------------------|
| <code>/sys/class/hwmon`</code> | none                                        | CPU/board temperature         |
| DHT11/DHT22                    | <code>adafruit-dht`</code>                  | Temperature + humidity        |
| BME680                         | <code>adafruit-circuitpython-bme680`</code> | Temp, humidity, pressure, gas |
| PIR motion                     | <code>gpiozero`</code>                      | Motion detection              |
| Door reed switch               | <code>gpiozero`</code>                      | Open/closed state             |

Config in `agent/config.ini`` under `[iot]``:

```
[iot] interval = 60 dht_type = DHT22 dht_pin = 4 bme680_i2c_addr = 0x77 pir_pin = 17 door_pin = 27
```

Setup: `python setup_agent.py <server_ip> <org_token>`` (same as standard agent), then `python iot_agent.py``.

### API: Sensor Readings (`api/routes/sensors.py``)

| Method | Path                                                  | Auth          | Description                                                                        |
|--------|-------------------------------------------------------|---------------|------------------------------------------------------------------------------------|
| POST   | <code>/api/sensors/&lt;device_id&gt;/readings`</code> | X-Agent-Token | Ingest sensor readings.<br>Body: <code>{sensor_type, value, unit, channel}`</code> |
| GET    | <code>/api/sensors/&lt;device_id&gt;/readings`</code> | JWT           | List readings for device (filterable by sensor_type, time range)                   |
| GET    | <code>/api/sensors/&lt;device_id&gt;/latest`</code>   | JWT           | Latest reading per sensor_type                                                     |

### MQTT Ingestion (`api/tasks/mqtt_tasks.py``)

Celery task subscribes to `{MQTT_TOPIC_PREFIX}/{#`}` when `MQTT_HOST`` is set. Topic convention: `{prefix}/{device_id}/sensors/{sensor_type}``. Payload: `{"value": <float>, "unit": "<str>", "channel": "<str|null>"}``. Writes to `SensorReading`` via the same path as the direct agent POST.

No-op when `MQTT_HOST`` is blank – no errors raised.

### SNMP Polling (`api/tasks/snmp_tasks.py``)

Per-device SNMP polling task. Community string stored in `Device.metadata.snmp_community`` (set via device Edit form in Admin). `SNMP_TIMEOUT`` env var controls per-request timeout (default 3s).

### Dashboard: IoT Sensors page (`dashboard/pages/18_IoT_Sensors.py``)

Shows sensor readings as time-series charts per sensor type. Displays latest value, unit, and channel for each sensor. Admin/Tech access only.

---

## ## 18. Test Suite

As of the 2026-09-24/25 test-coverage and security remediation passes (`audits/testing_audit.md``, `audits/security_audit.md``), every route file, every Celery task file, every Streamlit page, and every React page component has a dedicated test file. Full counts, all passing and stable across repeated runs:

| Suite                                    | Tests      | Coverage / notes                                                                                                                                                                                                                                                                                                        |
|------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>api/tests/`</code>                 | <b>490</b> | (487 passed + 3 xfailed)   72% line coverage ( <code>--cov-fail-under=70`</code> enforced in CI); <code>.coveragerc`</code> excludes <code>tests/`</code> , <code>migrations/`</code> , <code>seed.py`</code> from the denominator. All 26 <code>routes/*.py`</code> and all 20 <code>tasks/*.py`</code> files covered. |
| <code>dashboard/tests/`</code>           | <b>84</b>  | All 26 <code>pages/*.py`</code> files (incl. <code>_ticket_detail.py/`_client_ticket_detail.py`</code> ) via <code>streamlit.testing.v1.AppTest`</code> .                                                                                                                                                               |
| <code>frontend/`</code> (Vitest + RTL)   | <b>39</b>  | All 19 <code>src/pages/*.tsx`</code> files.                                                                                                                                                                                                                                                                             |
| <code>frontend/e2e/`</code> (Playwright) | <b>2</b>   | Verified against the live stack; installed but deliberately not wired into CI (needs Postgres+Redis+API+migrations+frontend running together – a distinct infra task).                                                                                                                                                  |
| <code>agent/tests/`</code>               | <b>19</b>  | Pure-logic tests (string/byte decoding, command-string building) – never call the Windows-only APIs, so this job runs on <code>windows-latest`</code> in CI even though the tests themselves are OS-agnostic.                                                                                                           |

One file per module is the convention going forward – when adding a new route or task file, add its test file in the same commit rather than letting coverage regress.

### ### Key testing infrastructure notes

- `**`api/tests/conftest.py`**` – session-scoped SQLite in-memory DB fixture shared across the whole ``api/`` suite. This sharing is an intentional speed tradeoff, not an oversight – but it means tests must not assume a clean DB and must use unique values (a UUID, not a hardcoded placeholder MAC/email/etc.) for anything that could collide with another test's leftover rows. Two real order-dependent collisions of exactly this kind were found and fixed during the coverage pass (see ``audits/testing_audit.md`` "bugs found," #2 and #9).
- `**`api/tasks/_app_singleton.py`**` – tests that exercise a Celery task against the test DB must prime ``tasks._app_singleton._app = app`` (not a per-task-module ``_app`` attribute).
- `**`api/.coveragerc`**` – omits ``tests/``, ``migrations/``, ``seed.py`` from the coverage denominator. Without it, CI's Codecov upload reported an inflated 54% (every test file is trivially ~100% "covered" by being imported) versus the real 44.4% at the time – a measurement artifact, not a real gap.
- `**`api/config.py`**` – ``TestConfig.BCRYPT_ROUNDS = 4`` (vs. 12 in production). Real bcrypt hashing in tests at production cost would make the suite unusably slow; this is a ~13x speedup and is why the whole ``api/`` suite runs in single-digit seconds, not tens.
- `**Streamlit `AppTest` gotchas**` (apply to every file in ``dashboard/tests/``):  
``st.page_link()/`st.switch_page()`` raise on an isolated page (``AppTest.from_file()`` can't resolve the multipage registry that only exists when Streamlit boots from ``app.py``) – patch both with  
``unittest.mock.patch("streamlit.page_link")/`patch("streamlit.switch_page")`` at the top of every test file. ``st.components.v1.html()`` content is completely opaque to ``AppTest`` (no ``html/`.components`` accessor) – assert on the mock call args instead of rendered output.
- `**`dashboard/utils/cached_calls.py`**` – never take the access token as an underscore-prefixed parameter on an ``@st.cache_data``-wrapped function; ``st.cache_data`` silently drops such parameters from the cache key, which caused the cross-session data leak documented in Chapter 22 below.

### ### Running tests

## API

```
cd api ; .\env\Scripts\Activate.ps1 ; pytest tests/ -v --cov=. --cov-config=.coveragerc  
--cov-report=term-missing
```

## Dashboard

```
cd dashboard ; .\env\Scripts\Activate.ps1 ; pytest tests/ -v
```

## Agent

```
cd agent ; .\env\Scripts\Activate.ps1 ; pytest tests/ -v
```

## Frontend (Vitest)

```
cd frontend ; npm test
```

## Frontend E2E (Playwright, against a running live stack)

```
cd frontend ; npx playwright test
```

```
`api/` tests need a running PostgreSQL + Redis (or the CI service containers).
`dashboard/`/`agent/` tests are pure Python and need neither. `conftest.py` force-sets
`ORG_REGISTRATION_TOKEN` and creates an in-memory test app via `create_app("testing")`.

### CI pipeline (`.github/workflows/ci.yml`)

Five jobs: `test-backend` (Postgres 16 + Redis 7 services, migrations, pytest+coverage,
`pip-audit` x3 requirements files), `test-dashboard` (`ubuntu-latest`), `test-agent`
(`windows-latest` – `pywin32`/`wmi` have no Linux wheel), `build-frontend` (`npm audit`,
`npm test`, `tsc --noEmit`, `npm run build`), `build-docker` (main branch only, gated on
`test-backend` + `build-frontend`). Both `pip-audit` and `npm audit` steps enforce (no
`continue-on-error`) as of the 2026-09-25 security remediation – see Chapter 20a.

---

## 19. Mobile Device Management

### Overview

Real management (lock, wipe, lost-mode, compliance reporting) for phones that cannot run
the RMM's Python agent. No custom agent binary is written or shipped – Android devices
run Google's own signed "Android Device Policy" app, provisioned during a
device-owner-initiated enrollment (QR code or link, never silent). The API talks to
Google's Android Management API server-side; the phone never talks to this RMM directly.
iOS is not implemented – see [iOS status](#ios-status) below.

Key files: `api/models/mdm_integration.py`, `api/utils/android_mgmt.py`,
`api/routes/mobile_mdm.py`, `api/tasks/mdm_tasks.py`, `api/utils/crypto.py`,
`dashboard/pages/19_Mobile_Enrollment.py`.

### Data model

**`MdmIntegration`** (`mdm_integrations`) – one row per bound Android Enterprise:
```

| Field                                 | Notes                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------|
| `customer_id`                         | Nullable – null means staff-wide, set means a dedicated per-customer binding         |
| `project_id` / `enterprise_id`        | Google Cloud project + `enterprises/{id}` resource name (set once binding completes) |
| `service_account_json_enc`            | Fernet-encrypted (`api/utils/crypto.py`), never written to disk                      |
| `apple_push_cert_enc` / `apple_topic` | Reserved placeholders for a future iOS phase – unused today                          |

**MobileEnrollment** (`mobile\_enrollments`) – one-to-zero-or-one with `Device`:

| Field                                                                                        | Notes                                                            |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| `device_id`                                                                                  | Nullable until Google confirms enrollment, unique                |
| `ownership_type`                                                                             | `byod` \ `corporate` – maps to Android's `allowPersonalUsage`    |
| `status`                                                                                     | `pending` \ `enrolled` \ `revoked` \ `wiped` \ `expired`         |
| `consent_given_by_user_id`, `consent_given_at`, `consent_text_version`, `consent_ip_address` | Real columns, not a JSON blob – this is the legal consent record |

`Device.is\_agentless` stays `True` for managed phones – they still run no RMM agent. Real action buttons in the dashboard gate on `device.mdm\_enrollment.status == "enrolled"`, not `is\_agentless`.

### API: `/api/mdm` (`api/routes/mobile\_mdm.py`)

| Method | Path                                                                                   | Auth                            | Description                                                                                                                  |
|--------|----------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| POST   | /api/mdm/integrations                                                                  | admin                           | Create integration stub                                                                                                      |
| POST   | /api/mdm/integrations/<id>/credentials                                                 | admin                           | Multipart upload of the Google service-account JSON                                                                          |
| POST   | /api/mdm/integrations/<id>/bind                                                        | admin                           | Step 1 of enterprise binding – returns `{signup_url}`                                                                        |
| GET    | /api/mdm/integrations/<id>/bind_callback                                               | none (state-nonce validated)    | Google's browser redirect target                                                                                             |
| PUT    | /api/mdm/integrations/<id>/policy                                                      | admin, technician               | Push a policy JSON to the bound enterprise                                                                                   |
| GET    | /api/mdm/available_integrations                                                        | any authenticated role          | Secret-free list scoped to the caller's customer – powers the Client Portal's self-enrollment dropdown                       |
| POST   | /api/mdm/enrollments                                                                   | admin, technician, client       | Create an enrollment token; `role=client` is server-forced to `ownership_type=byod` and requires `consent_acknowledged=true` |
| DELETE | /api/mdm/enrollments/<id>                                                              | scope-checked                   | Revoke – `RELINQUISH_OWNERSHIP` for BYOD, full removal for corporate                                                         |
| POST   | /api/mdm/devices/<id>/{lock,reboot,reset_password,wipe,start_lost_mode,stop_lost_mode} | scope-checked, per-command role | Issues an `issueCommand` – applies on the device's next check-in, not instantly                                              |

`wipe` is `admin`-only. `lock`/`start\_lost\_mode`/`stop\_lost\_mode` are the only commands a `role=client` may issue, and only on their own scoped device.

### Practical example: enroll and lock a company phone

## 1. Create the integration (once, per Google Cloud project)

```
curl -X POST http://localhost:5000/api/mdm/integrations \ -H "Authorization: Bearer $ADMIN_JWT" -H "Content-Type: application/json" \ -d '{"name": "Acme Fleet", "project_id": "acme-rmm-mdm"}'
```

→ `{"id": "e1f2...", "type": "android", "bound": false, ...}`

## 2. Upload the service-account JSON, then bind (opens Google's own signup page)

```
curl -X POST http://localhost:5000/api/mdm/integrations/e1f2.../credentials \ -H "Authorization: Bearer $ADMIN_JWT" -F "file=@service-account.json" curl -X POST http://localhost:5000/api/mdm/integrations/e1f2.../bind \ -H "Authorization: Bearer $ADMIN_JWT" -H "Content-Type: application/json" \ -d '{"callback_base_url": "https://your-tunnel.example.com"}'
```

→ `{"signup_url": "https://enterprise.google.com/signup/android/...&state=..."}`

Admin opens `signup_url` in a browser, completes Google's hosted flow,

Google redirects to `bind_callback` — `integration.enterprise_id` is now set.

## 3. Create a corporate enrollment and get the QR/link

```
curl -X POST http://localhost:5000/api/mdm/enrollments \ -H "Authorization: Bearer $ADMIN_JWT" -H "Content-Type: application/json" \ -d '{"mdm_integration_id": "e1f2...", "ownership_type": "corporate"}'
```

→ {

"enrollment\_id": "9a8b...",

"qr\_code\_json": "{...}", # render with the `qrcode` package

"enrollment\_link": "https://enterprise.google.com/android/enroll?et=ABC123",

"expires\_at": "2026-09-22T16:30:00+00:00"

}

Technician scans this at a factory-reset phone's welcome screen.

The next 5-minute beat tick  
(`tasks.mdm_tasks.sync_all_mdm_integrations`)

links the enrollment to a new Device row automatically.

## 4. Once enrolled, issue a real command

```
curl -X POST http://localhost:5000/api/mdm/devices/<device_id>/lock \ -H "Authorization: Bearer $ADMIN_JWT"
```

→ 202 {"message": "Command sent — applies when the device next checks in."}

### Practical example: minimal policy JSON

Pushed via ``PUT /api/mdm/integrations/<id>/policy``, stored as ``MdmIntegration.default_policy_name`` and applied to every device enrolled under it:

```
{ "policy_name": "default", "policy": { "passwordRequirements": { "passwordMinimumLength": 6 },  
  "cameraDisabled": false, "applications": [ { "packageName": "com.slack", "installType":
```

```
"FORCE_INSTALLED"}}}]}
```

App push/removal, kiosk mode, and per-app VPN are all additional fields on this same ``policy`` object – no new endpoint or table needed to extend it (see Google's Android Management API ``policies`` reference for the full schema, which evolves – re-verify field names before relying on any not shown here).

### Celery sync (``tasks.mdm_tasks``)

``sync_all_mdm_integrations`` runs every 5 minutes (``celery_app.py`` beat schedule), fanning out ``sync_mdm_integration.delay(id)`` per active, bound integration. Each sync call: lists devices via ``enterprises.devices.list``, matches new devices to still-pending ``MobileEnrollment`` rows, creates the corresponding ``Device`` row on first sight, and updates ``policy_compliant`/`hardware`` fields on every pass – the same shape as ``tasks.psa_tasks.sync_all_psa_integrations``.

### Consent enforcement

Enrollment consent is checked server-side, not just hidden in the UI: ``create_enrollment`` in ``mobile_mdm.py`` forces ``ownership_type="byod"`` for any ``role=client`` caller regardless of the request body, requires ``consent_acknowledged=true`` for BYOD, and 404s if a client tries to enroll into an integration scoped to a different customer's ``customer_id``. Every consent event and every issued command writes an ``AuditLog`` row (``resource_type="mobile_device"``).

### iOS status

Not implemented. Real Apple MDM requires an APNs push certificate, which Apple only counter-signs for organizations enrolled in Apple Business Manager or for already vendor-signed MDM servers (Fleet, MicroMDM/NanoMDM) – a business/account decision outside this codebase. ``MdmIntegration.type`` already accepts ``"apple"`` and the ``apple_push_cert_enc`/`apple_topic`` columns exist, so this phase needs no schema migration when the decision is made – only a new branch in ``MdmIntegration.get_client()``, which currently raises ``NotImplementedError`` for ``type="apple"``.

## 20. API & Token Usage Monitoring

### Overview

Superadmin-only backend auditing/reporting for API call volume and AI token consumption, built after usage got unexpectedly high with no visibility into which service, feature, or user was responsible. Strictly gated to ``role == "superadmin"`` – ``admin`` gets 403/sees nothing, unlike every other admin surface in this app.

Key files: ``api/models/usage.py``, ``api/utils/usage_tracker.py``, ``api/routes/usage.py``, ``api/tasks/usage_tasks.py``, ``dashboard/pages/22_Usage_Monitoring.py``, ``frontend/src/pages/UsageMonitoringPage.tsx``.

### Data model

`**`ApiUsageEvent`**` (``api_usage_events``) – one row per instrumented call:

| Field | Notes |
|-------|-------|
| ----- | ----- |

```
| `service` | `ai_assistant`, `stripe`, `psa_connectwise`, `psa_autotask`, `android_mdm`,
`email_imap`, `network_scan`, `webhook_slack`/`webhook_teams`/`webhook_generic`, `smtp` |
| `feature` | Sub-label – dashboard page name for `ai_assistant`, action name (e.g.
`get_companies`) for integrations |
| `user_id` | Nullable – null for Celery/system-triggered calls |
| `input_tokens` / `output_tokens` / `estimated_cost_usd` | Only populated for
`ai_assistant`; cost is a configurable estimate, not billing data |
| `status` / `status_code` / `latency_ms` / `error_message` | `error_message` is a
truncated exception type/message only – never a raw response body, header, or secret |
```

**ApiUsageHourly** (`api\_usage\_hourly`) – durable rollup of internal Flask API request volume (`bucket\_start`, `endpoint`, `method`, `request\_count`, `error\_count`, `total\_latency\_ms`). Too high-volume to log one `ApiUsageEvent` row per request, so `utils/usage\_tracker.py::record\_internal\_api\_call` increments a Redis hash (`rmm:usage:hourly:{YYYYMMDDHH}`, 8-day TTL) on every request via `app.py`'s existing `\_log\_request` `after\_request` hook, and `tasks.usage\_tasks.persist\_hourly\_usage\_rollup` (hourly beat) upserts the previous completed hour's counters into this table before the Redis TTL expires.

**UsageAlertConfig** (`usage\_alert\_config`) – singleton row (`id="default"`):  
`is\_enabled`, `spike\_multiplier` (default 3.0), `notification\_channels` (same JSON shape as `AlertRule.notification\_channels`). Deliberately its **own** table, not an `AlertRule` row – an `AlertRule` row would be visible to `admin`/`technician` via the Alerts page's "Alert Rules" tab, leaking this feature's existence/config to roles that must not see it.

Migration: `q8r9s0t1u2v3` (revises `p7q8r9s0t1u2`).

### API: `/api/admin/usage` (`api/routes/usage.py`)

| Method    | Path                                                                   | Auth            | Description                                                                         |
|-----------|------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------|
| GET       | /api/admin/usage/summary?range=today\ 7d\ 30d                          | superadmin only | Per-service totals (calls, tokens, estimated cost, error rate) + live anomaly flags |
| GET       | /api/admin/usage/timeseries?range=&service=&metric=calls\ tokens\ cost | superadmin only | Bucketed series for charts                                                          |
| GET       | /api/admin/usage/by-feature?range=&service=                            | superadmin only | Top features/pages/users breakdown                                                  |
| GET       | /api/admin/usage/events?service=&status=&page=                         | superadmin only | Paginated raw event drill-down                                                      |
| GET / PUT | /api/admin/usage/alert-config                                          | superadmin only | Read/update the spike-alert config                                                  |

`\_require\_superadmin()` in this file has **no** admin bypass – unlike every other route file's `\_require\_role()`, which lets `superadmin` bypass an explicit admin/technician list. Here even `admin` is excluded outright.

### Practical example: check today's AI Assistant spend

```
curl http://localhost:5000/api/admin/usage/summary?range=today -H "Authorization: Bearer $SUPERADMIN_JWT"
```

→ {

"range": "today",

"services": [

{"service": "ai\_assistant", "calls": 214, "input\_tokens": 61000,

"output\_tokens": 38500, "estimated\_cost\_usd": 0.2535, "error\_count": 1, "error\_rate": 0.005},

{"service": "internal\_api", "calls": 18420, "input\_tokens": 0, "output\_tokens": 0,

"estimated\_cost\_usd": 0, "error\_count": 32, "error\_rate": 0.002}

],

"totals": {"calls": 18634, "tokens": 99500, "estimated\_cost\_usd": 0.2535,

"error\_count": 33, "error\_rate": 0.0018},

"anomalies": []

}

### ### Spike detection and alerting

``utils.usage_tracker.compute_anomalies(spike_multiplier)`` compares the most recently **completed** hour's per-service count against the trailing-7-day average for that same hour-of-day, flagging anything over the multiplier (default 3.0×). This same function powers both the live banner on ``22_Usage_Monitoring.py``/``UsageMonitoringPage.tsx`` and the hourly ``tasks.usage_tasks.detect_usage_anomaly`` beat task, which – only if ``UsageAlertConfig.is_enabled`` – calls ``send_alert_notification``/``dispatch_alert_webhooks`` **directly** with a synthetic ``device_hostname="System / API Usage"`` identity. No ``Alert``/``AlertRule`` row is created: ``Alert.device_id`` is ``NOT NULL`` and the whole ``evaluate_all_rules`` batch-query machinery is written around real online devices, so extending it for a non-device concept was judged not worth the schema migration – reusing the already-decoupled notification primitives directly was the cleaner path.

### ### Retention and reports

``tasks.maintenance_tasks.prune_old_data`` (existing daily beat, extended) deletes ``ApiUsageEvent`` rows older than 90 days and ``ApiUsageHourly`` rows older than 180 days. A new ``api_usage`` Reports template type (``tasks.report_tasks._api_usage_summary``) produces a per-day-per-service CSV via the existing async Celery report pipeline – ``reports.py`` hides this template from list/generate/get for any non-superadmin caller, same as the API endpoints above.

### ### What's deliberately NOT tracked

Per-ping/per-host detail during a network scan (only one event per scan run, with host count in ``feature``), and the six legacy per-function SMTP call sites in ``utils/notifications.py`` that predate the shared ``_smtp_send`` helper (only ``_smtp_send`` itself and ``billing.py``'s invoice-email block are instrumented) – both would add event volume disproportionate to their value for finding the cause of high usage.

---

## ## 21. Consistent List Pagination

### ### Why

``audits/code_duplication_audit.md`` (Finding N2) found the same ~15-line pagination block – parse ``page``/``per_page`` query args, clamp to a max, run ``.paginate()``, build the response dict – hand-copied across 9+ list endpoints, with two real inconsistencies as a result: some endpoints returned a ``"pages"`` key and some didn't, and ``admin.py::list_users`` returned its collection under key ``"users"`` while every other list endpoint used ``"items"`` – a real, live bug, since the React Admin page read ``data.items`` and was silently getting an empty list.

### ### ``api/utils/pagination.py``

`def paginated_response(query, serialize, *, order_by=None, default_per_page=50, max_per_page=200, items_key="items"):` `query` — an unfiltered, unordered SQLAlchemy Query `serialize` — callable applied to each row (usually ``.to_dict``) `order_by` — optional column/expression to order by before paginating

Reads page/per\_page from flask.request.args; per\_page is clamped to [1, max\_per\_page]. Returns a dict: {items\_key: [...], "total": N, "page": N, "pages": N}. ""

```
### Call sites
```

```
Wired into: `alerts.py::list_rules`, `automation.py::list_profiles`/`list_runs`,  
`customers.py::list_customers` (kept its own `default_per_page=20, max_per_page=100` – a  
deliberately different default for a smaller, more-often-fully-browsed collection),  
`customers.py::list_groups`, `patches.py::list_patches`, `scripts.py::list_runs`,  
`admin.py::list_users` (the fix for the `"users"`/`"items"` key bug above – the matching  
dashboard-side fix was in `dashboard/pages/02_Tickets.py`, `10_Admin.py`, and  
`_ticket_detail.py`, all of which had a stray `.get("users", [])` changed to  
`.get("items", [])`).
```

```
**Deliberately not converted:** `devices.py`'s list endpoint (its pagination is entangled  
with the platform-filter/cache-key logic and didn't cleanly decompose), and  
`alerts.py::list_alerts`/`usage.py::events` (both needed a slightly different shape and  
got a smaller inline fix instead of forcing the shared helper).
```

```
### Adding a new paginated endpoint
```

```
from utils.pagination import paginated_response
```

```
@bp.route("/", methods=["GET"]) @jwt_required() def list_things(): query =  
Thing.query.filter_by(customer_id=customer_id) return jsonify(paginated_response(query, lambda t:  
t.to_dict(), order_by=Thing.created_at.desc()))
```

```
Use this instead of hand-rolling `.paginate()` again unless your endpoint's shape  
genuinely doesn't fit one of the two documented exceptions above.
```

```
---
```

## ## 22. Security Audit Remediation & Dependency Hygiene

```
`audits/security_audit.md` (2026-09-25) covered injection, cryptography, secrets  
handling, dependency/supply-chain posture, and configuration hardening. All 7 scored  
findings are fixed and verified – API suite 490/490, dashboard 84/84, frontend 39/39  
Vitest + 2/2 Playwright, agent 19/19, coverage unchanged at 72% (fixes to existing code  
paths only, no new production surface).
```

### ### 22.1 Stored XSS – `dashboard/pages/04\_Devices.py` (Finding I1, 8/10 – the standout finding)

```
Two `st.markdown(f""...""`, unsafe_allow_html=True)` blocks rendered `hostname`,  
`ip_address`, `os_name`, `os_version`, `platform`, `cpu_model` directly into an f-string  
without the `esc()` helper this codebase uses everywhere else for the same purpose.  
Exploitable via two different write paths that don't require an authenticated dashboard  
session: the page's own inline agentless-device "Edit" form (any admin/technician), and a  
raw `POST /api/agents/register` call authenticated with nothing but the org registration  
token – a value this same repo's changelog documents was once leaked into git history  
(see the 2026-09-21 diagnostic entry). The payload fires in the browser of whoever next  
views that device.
```

```
**Fix:** wrap `esc()` around all 6 fields in both render blocks. Regression test:  
`dashboard/tests/test_devices_page.py::test_hostname_is_html_escaped` asserts a `<img`  
onerror=...>` payload renders as escaped HTML entities, not live markup.
```

### 22.2 ``api/utils/crypto.py`` hardening (Findings S1, C1)

- **\*\*No longer fails open\*\*** (S1, 6/10): ``encrypt_cred`/`decrypt_cred`` previously had ``except Exception: return value`` – any failure (a corrupted ciphertext, a rotated key) silently returned the raw input, which for ``decrypt_cred`` meant potentially treating still-encrypted ciphertext as a usable plaintext credential. Removed; failures now propagate to ``api/app.py``'s existing app-wide ``@app.errorhandler(Exception)``, which already returns a generic 500 with server-side ``exc_info=True`` logging – the same convention used everywhere else in this codebase.
- **\*\*Domain-separated key\*\*** (C1, 4/10): ``_fernet()`` now derives its key from ``SHA256("rmm-cred-encryption:" + SECRET_KEY)`` instead of ``SHA256(SECRET_KEY)`` alone, so this key is cryptographically distinct from any other future use of ``SECRET_KEY``.
- No data migration was needed for either change (verified zero pre-existing encrypted rows in the live DB before the key-derivation change). ``api/tests/test_crypto.py`` (5 tests) covers the round-trip and the now-required raise-on-failure behavior.

### 22.3 MFA secret encryption (Finding S2, 6/10)

``User.mfa_secret`` (the TOTP shared secret) was stored in plaintext. ``api/models/user.py`` gained ``set_mfa_secret(secret)`/`get_mfa_secret()``, both routing through ``utils.crypto``; all 3 call sites in ``routes/auth.py`` (MFA setup, enable, login verification) updated to use them instead of touching the column directly. Verified live against the real Postgres DB: the column now holds a ``gAAAAA...`` Fernet token, and the full setup → enable → login round trip was re-tested end-to-end afterward.

### 22.4 Password reset tokens are now single-use (Finding A1, 5/10)

A reset token is a stateless JWT (``purpose: "password_reset"``, 1-hour expiry) – correctly scoped, but previously replayable for its whole validity window after first use, since nothing was checked against it having already been consumed.

## api/routes/auth.py::password\_reset\_confirm — after decoding, before setting the new password

```
if user.password_changed_at: issued_at = datetime.fromtimestamp(decoded.get("iat", 0), tz=timezone.utc)  
changed_at = user.password_changed_at if changed_at.tzinfo is None: # SQLite (and some Postgres  
driver configs) changed_at = changed_at.replace(tzinfo=timezone.utc) # return this column naive if  
issued_at < changed_at: return jsonify({"error": "Invalid or expired reset link"}), 400
```